

SPECIFICATIONS FOR EXPRESS JOBDECK FILE FORMAT

This manual applies to the following
JEOL products:

JBX-5000LS,	JBX-5000LS/E
JBX-6000FS,	JBX-6000FS/E
JBX-9000MV,	JBX-9000MV/II
JBX-9300FS,	JBX-3030MV
JBX-3040MV,	JBX-3050MV
JBX-6300 series	JBX-5500 series

For the proper use of the instrument, be sure to read this instruction manual. Even after you read it, please keep the manual on hand so that you can consult it whenever necessary.

SPECIFICATIONS FOR EXPRESS JOBDECK FILE FORMAT

**Please be sure to read this instruction manual carefully,
and fully understand its contents prior to the operation
or maintenance for the proper use of the instrument.**

NOTICE

- The following actions must be avoided without prior written permission from JEOL Ltd. or its subsidiary company responsible for the subject (hereinafter referred to as "JEOL"): modifying the instrument; attaching products other than those supplied by JEOL; repairing the instrument, components and parts that have failed, such as replacing pipes in the cooling water system, without consulting your JEOL service office; and adjusting the specified parts that only field service technicians employed or authorized by JEOL are allowed to adjust, such as bolts or regulators which need to be tightened with appropriate torque. Doing any of the above might result in instrument failure and/or a serious accident. If any such modification, attachment, replacement or adjustment is made, all the stipulated warranties and preventative maintenances and/or services contracted by JEOL or its affiliated company or authorized representative will be void.
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NOTATIONAL CONVENTIONS AND GLOSSARY

■ Examples for general notations

– CAUTION – :	Important precautions for use, which, if not followed, may result in damage to or problems with the device itself.
 :	Additional points to remember regarding the operation.
 :	A reference to another section, chapter or manual.
1, 2, 3 :	Numbers indicate a series of operations that achieve a task.
 :	A diamond indicates a single operation that achieves a task.
File :	The names of menus, commands, or parameters displayed on the screen are denoted with bold letters.
File–Exit :	Selecting a menu item from a pulldown menu is denoted by linking the menu and the item with a dash (–). For example, File–Exit means selecting Exit from the File menu.
 :	Keys on the keyboard are denoted by enclosing their names in a box.

■ Examples for mouse terminology

Mouse pointer:	A mark, displayed on the screen, which moves following the movement of the mouse. It is used to specify a menu item, command, parameter value, and other items. Its shape changes according to the situation.
Click:	To press and release the left mouse button.
Right-click:	To press and release the right mouse button.
Double-click:	To press and release the left mouse button twice quickly.
Drag:	To hold down the left mouse button while moving the mouse.

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SAFETY PRECAUTIONS

Before using the instrument, be sure to read this instruction manual, and upon fully understanding its contents, use the instrument properly. This section, “Safety Precautions”, contains important information about safety.

- **The safety indications and their meanings are as follows:**

	DANGER:	An imminently hazardous situation which, if not avoided, will result in death or serious injury.
	WARNING:	A potentially hazardous situation which, if not avoided, could result in death or serious injury.
	CAUTION:	A potentially hazardous situation which, if not avoided, may result in minor or moderate injury, or a situation that could result in serious damage to facilities or acquired data.

- **Labels bearing the following symbols are attached to dangerous locations on the instrument. Do not touch any of these locations with your hands or anything else.**



Examples of symbols

- Use the instrument properly within the scope of the purpose and usage described in its brochures and manuals.
- Never remove or disable protective parts, safety devices, or other safety features.
- Never make modifications that include disassembling the instrument or installing substitute parts.
- Never disconnect the grounding wire or move it from the prescribed position. Failure to follow this instruction could result in electric shock.
- When you dispose of the instrument or liquid or other waste, follow all applicable laws and regulations, and dispose of it in a proper manner without polluting the environment.
- Be sure to read the “Safety Precautions” section of the manuals for the accessories attached to or built into the instrument.
- If anything is unclear, please contact your local JEOL service office.

1 GENERAL

This manual describes files used in a JEOL electron beam lithography system* and in which jobdecks (electron-beam writing schedules) for the EXPRESS job schedule preparation system for e-beam writing are written. This manual also presents the format specifications of commands and parameters used in the files.

EXPRESS (EXposure PREparation System for Scheduling), a software system, is designed for JEOL e-beam lithography systems.

A jobdeck is a series of command parameter lines that controls the e-beam writing process, including designation of the types and sizes of substrates, arrangement of chip patterns and character strings on the substrates, and scheduling of the jobs involved in pattern writing.

A file in which a jobdeck is written is called a jobdeck file.

Jobdeck files are primarily designed by the user, who specifies the positions of various marks, chip patterns, and character strings on substrates. Jobdeck files are easy to create, without extensive knowledge of the control system of the e-beam lithography system.

Jobdeck files support writing of multilayer devices. Since each file contains one common block and one or more layer blocks, the user specifies parameters common to all layers in the common block, and parameters specific to individual layers in each layer block. Usually, the common block concerns positions of various marks, and positions for writing chip patterns and character strings on the substrate, while the layer blocks concern the names of chip patterns to be written and the contents of character strings.

In electron-beam writing schedules, some of the commands issued in jobdeck files can also be issued in schedule files. Those issued in schedule files have priority over these commands in jobdeck files.

Jobdeck files can be created and edited by using the file editor utility of various computer systems. Jobdeck commands are expressed in the ASCII using numerals, uppercase letters, and several special symbols.

* JEOL electron-beam lithography systems are of various types. Unless otherwise noted, the model name JBX-6000FS written in the text or footnotes also applies to models such as the JBX-5DII, JBX-5FE, and JBX-5000LS. Similarly, the JBX-6000FS/E also applies to models such as the JBX-5000LS/E, the JBX-9000MV also applies to the JBX-9000MV/II, and the JBX-3030MV also applies to the JBX-3040MV and the JBX-3050MV, the JBX-6300series also applies to the JBX-6300xx (xx: Last characters following JBX-6300.), and the JBX-5500series also applies to the JBX-5500xx (xx: Last characters following JBX-5500.) Electron Beam Lithography System.

2 FORMAT

2.1 File Format

A jobdeck file is created on a magnetic recording medium such as a disk or tape and has the following format:

- Sequential
- Formatted
- Record size: Variable length (up to 255 bytes/record)
- Record attributes: Carriage return/carriage control

The name of a file name is limited as follows.

“(jobdeck file name).JDF” or “(jobdeck file name).jdf”

 (jobdeck file name) is a character string consisting of any alphabets, numbers, and/or symbols.

A jobdeck file consists of a common block and one or more layer blocks. The common block contains the parameters common to all layers within the jobdeck file, and each layer block contains parameters specific to the individual layer.

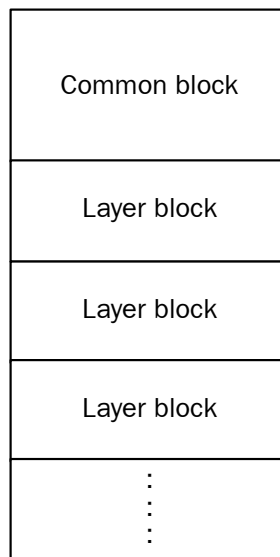


Fig.1 Jobdeck file format

One record in a jobdeck file is equivalent to one physical record in which command parameter lines of a jobdeck are stored. These command parameter lines can be divided into two or more physical records, called logical lines. A jobdeck file consists of multiple logical lines.

2.2 Command Format

Various parameters in a jobdeck are specified by using commands. Jobdeck commands are hierarchically constructed as shown below.

Table 1 Definition of Symbols

Symbol	Description
=	Item on the left side is composed of elements on the right side.
+	Indicates sequence of commands.
[]	The elements in square brackets [] can be used zero or more times.
[]	The elements in thick brackets [] can be used zero or one time(s). Although the same element can be designated two or more times, only the last one is in effect.
[]	The elements in thin brackets [] can be used one or more times.
{ }	One of the elements separated by vertical bars within braces { } must be selected.
{ / }	The elements separated by slashes / within braces { } can be used in an arbitrary order.
< >	The element within angle brackets < > represents other commands.

Jobdeck file = <Common block> + [<Layer block>] + END

<Common block> = JOB + [OPTION] + [DEVINF] + [<Global mark definition>
+ [<Plate mark definition>] + [<Writing path definition>
+ [{<Writing pattern definition>
/ <Writing character string definition>}]

<Global mark definition> = GLMPOS + [GLMP] + [GLMQRS]

<Plate mark definition> = PLMPOS + [PLMARK]

<Writing path definition> = PATH + [SCALE]
+ [{<Chip array definition>
/ <Character string position definition>}] + PEND

<Chip array definition> = ARRAY + [<Chip mark definition>
+ [{ASSIGN/SKIP}] + AEND

<Chip mark definition> = CHMPOS + [CHMARK]

<Character string position definition> = CHAR

<Writing pattern definition> = P + [{[SPPRM]/[STGPRM]}]

<Writing character string definition> = C

<Layer block> = LAYER + [CIRCLE] + [CANCEL] + [MIRROR] + [SCALE]
+ [{<Writing pattern definition>
/ <Writing character string definition>}]
+ { [RESIST]/[SHOT]/[MODULAT] *
/[EOS]/[DEFOCUS] **
/[RESTYP]/[STDCUR]/[SCALEB]/[OBJAPT]/[PWIDTH] *** }

* RESIST, SHOT, and MODULAT are generically called "writing condition definitions."

** EOS and DEFOCUS are generically called "hardware condition definitions."

*** RESTYP, STDCUR, SCALEB, OBJAPT, and PWIDTH are generically called "writing condition assignments."

3 OUTLINE OF COMMANDS AND CONTROL CHARACTERS

3.1 Outline of Commands

The jobdeck commands are shown in the tables below.

Table 2 Jobdeck commands (Common block)

No.	Command	Description	Remarks
(1)	JOB	Declares the beginning of a jobdeck; specifies its name, the type and size of substrate, and diameter of circular cut.	Required
(2)	OPTION	Specifies optional parameters common to all layers.	Optional
(3)	DEVINF	Specifies character strings concerning devices common to all layers.	Optional
(4)	GLMPOS	Specifies writing in the direct writing (DW) mode or in the phase shift mask pattern writing (PSM) mode, and specifies the positions of global marks on substrate.	Required in DW and PSM modes
(5)	GLMP	Specifies the shape of global mark P.	Optional in DW and PSM modes
(6)	GLMQRS	Specifies the shape of global marks Q, R, S.	Optional in DW and PSM modes
(7)	PLMPOS	Specifies the positions of plate marks on substrate for writing in the phase shift mask pattern writing (PSM) mode.	Required in PSM mode
(8)	PLMARK	Specifies the shapes of plate marks on substrate for writing in the phase shift mask pattern writing (PSM) mode.	Optional in PSM mode
(9)	PATH	Specifies the beginning of a writing path definition.	Required
(10)	SCALE	Specifies execution of fine pitch control writing and the magnification or reduction ratio for that writing.	Optional
(11)	ARRAY	Specifies the beginning of a chip array definition, and writing position of the array of chips.	Optional
(12)	CHMPOS	Specifies positions of chip marks on substrate in the direct writing (DW) mode.	Optional in DW mode
(13)	CHMARK	Specifies the shapes of chip marks.	Optional in DW mode
(14)	ASSIGN	Specifies allocation of chips and arrays of chips to arbitrary array points on substrate.	Optional
(15)	SKIP	Skips chips and arrays of chips allocated to arbitrary array points on substrate.	Optional
(16)	AEND	Specifies the end of a chip array definition.	Optional
(17)	CHAR	Specifies the writing position and attributes of a character string.	Optional
(18)	PEND*	Specifies the end of a writing path definition.	Required
(19)	P	Specifies writing of a chip at the array position assigned by a	Optional

* It must be a pair with the preceding PATH command.

No.	Command	Description	Remarks
		pseudo pattern number, the shift from the array position, and a specific writing mode for each pattern writing.	
(20)	SPPRM	Specifies shot partition parameters for each pattern.	Optional
(21)	STGPRM	Specifies the method for moving the substrate stage during pattern writing.	Optional, available in the 3050MV and later models
(22)	C	Specifies the contents of a character string.	Optional

Table 3 Jobdeck commands (Layer block)

No.	Command	Description	Remarks
(1)	LAYER	Declares the beginning of a layer block; specifies the layer number and name as a character string, and execution of writing in the phase shift mask pattern writing (PSM) mode.	Required
(2)	CIRCLE	Specifies cutting of a circle for chip array, and its diameter and attributes.	Optional
(3)	CANCEL	Cancels writing of the character strings that represent the date, the time, the machine identification name, and the device type.	Optional
(4)	MIRROR	Specifies execution of axis inversion of writing position of chips and character strings, and its attributes.	Optional
(5)	SCALE	Specifies execution of fine pitch control writing and the magnification or reduction ratio for that writing.	Optional
(6)	P	Specifies writing of a chip at the array position assigned by a pseudo pattern number, the shift from the array position, and a specific writing mode for each pattern writing.	Optional
(7)	SPPRM	Specifies shot partition parameters for each pattern.	Optional
(8)	STGPRM	Specifies the method for moving the substrate stage during pattern writing.	Optional, available in the 3050MV and later models
(9)	C	Specifies the contents of a character string.	Optional
(10)	RESIST	Specifies resist sensitivities used in pattern writing.	Optional
(11)	SHOT	Specifies time for each e-beam shot.	Optional
(12)	MODULAT	Specifies the content of a shot modulation table.	Optional
(13)	EOS	Specifies both the EOS mode and table number to be used in e-beam writing. (To be used exclusively for JBX-6000FS, JBX-6000FS/E, JBX-6300 series, JBX-5500 series)	Optional
(14)	DEFOCUS	Specifies the correction level of the objective lens.	Optional
(15)	SCALEB	Specifies offset amount (bias value) of magnification or reduction ratio in fine pitch control writing.	Optional
(16)	OBJAPT	Specifies the objective lens movable aperture number.	Optional
(17)	STDCUR	Specifies beam current.	Optional

No.	Command	Description	Remarks
(18)	RESTYP	Specifies type and name of resist used in pattern writing.	Optional
(19)	PWIDTH	Specifies minimum line width of pattern or CD (critical dimension) pattern width.	Optional

Table 4 Jobdeck commands (Footer)

No.	Command	Description	Remarks
(1)	END	Declares the end of the jobdeck.	Required

3.2 Outline of Control Characters

The control characters that can be used in jobdeck files are shown in the table below.

Table 5 List of control characters

No.	Symbol	Description	Remarks
(1)	–	Declares that this line continues from the previous command parameter line.	Optional
(2)	=	Declares that the command used in the preceding line is also valid in this line.	Optional
(3)	;	Declares that character string that follows this semicolon is a comment.	Optional
(4)	@	Specifies a position where a file is to be inserted.	Optional

4 FORMAT AND FUNCTION OF COMMANDS AND CONTROL CHARACTERS

4.1 Format of Jobdeck Command

Typical formats of jobdeck commands are shown below.

①	②	③	④
⑤			

① **Label**

Identifies the ARRAY command allotted by the ASSIGN command (described later).

② **△**

Delimiter that consists of one or more consecutive spaces or tabs; it separates the label and the table from the command, and the command from the parameter list.

③ **Command**

Command to be given; enter here one of the command names or control characters listed in Chapter 3: OUTLINE OF COMMANDS AND CONTROL CHARACTERS.

④ **Parameter list**

One or more parameters specific to the command given; if a parameter list consists of multiple parameters, separate adjacent parameters with a comma (,).

⑤ **Table**

Identifies the MODULAT command allotted by the ASSIGN command (described later).

 Although one physical record can contain up to 255 characters, the system ignores all characters after the 132nd in one line. When you wish to write a command parameter line that exceeds 132 characters, put a hyphen (–) in the first column of the succeeding line to declare continuation of the command parameter line and enter a delimiter (△); then write the rest of the parameter list.

 A carriage return code (CR/LF) or a comment code (;) ends a command parameter line. However, if the command parameter line continues to the next line, the system will ignore it.

4.2 Common Block Commands

4.2.1 JOB

(Required)

Format:					
JOB	【/W】	【'j'】	d1	【,d2【a】】	【,b】
	①		②	③ ④ ⑤	⑥
Example:					
JOB		'JBNAME'	5		(1)
JOB/W		'JBNAME'	5,4.5		(2)

● Descriptions of the examples

- (1) This statement indicates that the name of the jobdeck is JBNAME and specifies that the substrate to be used is a 5-inch mask blank.
- (2) This statement indicates that the name of the jobdeck is JBNAME and specifies that the substrate to be used is a 5-inch wafer with a 4.5-inch diameter circular cutout of chip array.

For the chip array's circular cutout attribute (inside or outside of the circle) and the diameter unit (inch or millimeter), the inch is specified by default. The default writing attribute is the inside of the circle.

● Description of the command

The JOB command starts a jobdeck, and specifies the shape of the substrate, jobdeck name, substrate size, the diameter for the circular cutout, and location of chip array (inside or outside of circle). The command is always put at the beginning of a series of jobdeck commands and must not be omitted.

① /W

Shape of substrate; specify this for e-beam writing on round substrates such as wafers, but omit it for writing on rectangular substrates such as mask blanks for mask or reticle writing.

② j

Jobdeck name; it is recommended that the name agree with the jobdeck file name. This name is only a comment and will not affect system operation.

③ d1

Substrate size.

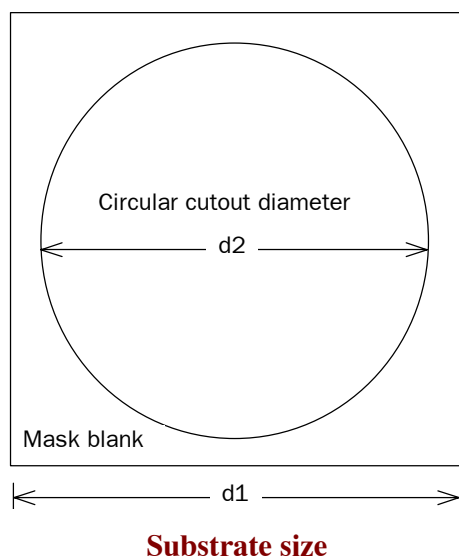
④ d2

Attributes designated by diameter of circular cutout of chip array and sign of d2.

d2	Attribute
> 0	Writes chips that lie on the periphery of the circle.
< 0	Does not write chips on the periphery of the circle.
= 0	No circular cutting.

 When the specification of d2 is omitted, the system will use the same value as d1 for writing on round substrates, but circular cutout for the array will not be performed when writing on rectangular substrates.

The parameters d1 and d2 are illustrated in the following figure.



⑤ a
Units of circular cutout diameter.

a	Unit
I	Inch (default)
M	mm

⑥ b
Attributes of circular cutout of chip array.

b	Attribute
I	Inside of circle (default)
O	Outside of circle

4.2.2 OPTION

(Optional)

Format:
OPTION [t] [, [i] [, [p] [, [c] [, sf]]]]
 ① ② ③ ④ ⑤

Example:
OPTION T,M

● Description of the example

Date and time of writing, and the machine name (the identification name of the e-beam lithography system used) will be written on the substrate.

● Description of the command

The OPTION command specifies optional parameters common to all layers. Optional parameters include the date and time of writing, the machine name for identification, and the mirroring of chip writing positions and character strings with respect to the X or Y axis.

① t

Specifies whether to write character strings indicating the date and time of writing for all layers contained in a specific device. The position and attributes of the character strings are specified by the CHAR command (CHAR/T).

t	Description
	Date and time are not written (default).
T	Date and time are written.

② i

Specifies whether to write character strings indicating the machine used for writing for all layers contained in a specific device. The position and attributes of the character strings are specified by the CHAR command (CHAR/M).

i	Description
	Machine name is not written (default).
M	Machine name is written.

③ p

Specifies whether to perform the mirroring of chip writing positions with respect to the X or Y axis for all layers contained in a specific device.

p	Description
	No mirroring (default)
X	X mirroring
Y	Y mirroring

④ c

Specifies whether to perform the mirroring of character strings with respect to the X or Y axis for all layers contained in a specific device.

 When writing position is mirrored in mask writing, the attributes of characters (normal or mirrored) will also change.

p	Description
	No mirroring (default)
X	X mirroring
Y	Y mirroring

⑤ sf

Specifies the base direction for a character string (that is, the direction of a normal character) for all layers contained in this device.

sf	Description
0	For horizontal characters, if the Y coordinate value of the writing position of a character string in the substrate center coordinate system is less than or equal to zero, the normal characters are oriented so that the direction in which Y coordinate values increase is upward; if this Y coordinate value is greater than zero, the normal characters are oriented so that the direction in which Y coordinate values decrease is upward. For vertical characters, if the X coordinate value of the writing position of a character string in the same coordinate system is less than or equal to zero, the normal characters are oriented so that the direction in which X coordinate values increase is upward; if this X coordinate value is greater than zero, the normal characters are oriented so that the direction in which X coordinate values decrease is upward. (Default)
1	Normal characters are oriented so that the direction in which the Y coordinate values in the substrate center coordinate system increase is upward.

4.2.3 DEVINF

(Optional)

Format:	
DEVINF	'd' ①
Example:	
DEVINF	'JEOL'

- **Description of the example**

The device name JEOL will be written on the substrate.

- **Description of the command**

The DEVINF command specifies writing a character string indicating the type of device common to all layers. If this command is not given, the system will not write the device type.

① d

Character string indicating the device type. The character string representing the device name is written on the substrate. The position and attributes of the character string are specified by the CHAR command (CHAR/D).

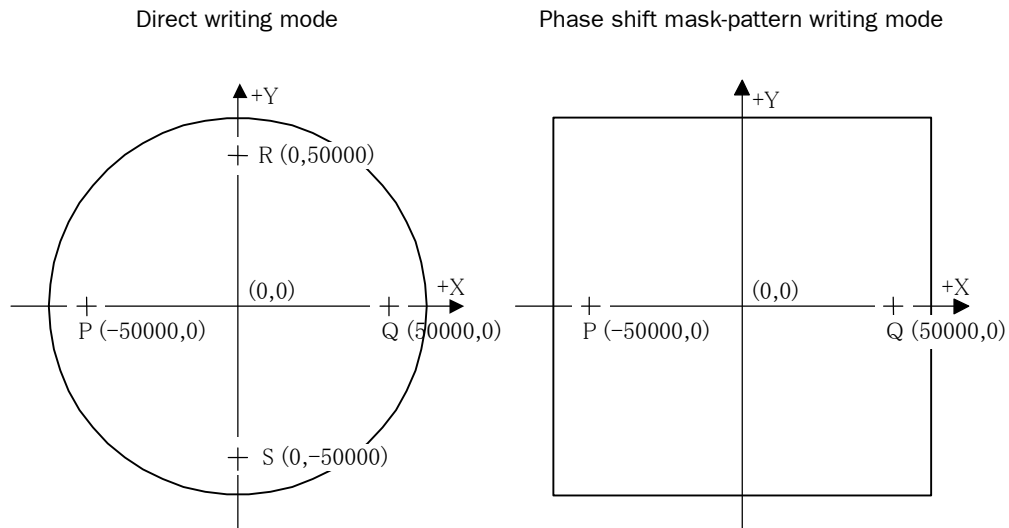
4.2.4 GLMPOS

(Required in both direct writing and phase shift mask pattern writing)

Format:	
GLMPOS	P=(px,py){,Q=(qx,qy)/,
	① ② ③ ④
–	R=(rx,ry)/,S=(sx,sy)}
	⑤ ⑥ ⑦ ⑧
Example:	
GLMPOS	P=(-50000.0,0.0),Q=(50000.0,0.0),
–	R=(0.0,50000.0),S=(0.0,-50000.0)

● Description of the example

These statements specify not only the positions of global marks P, Q, R, and S, but also execution of e-beam writing in the direct writing (DW) mode or the phase shift mask pattern writing (PSM) mode. The positions of global marks P, Q, R, and S on the substrate in the DW mode, and P and Q in the PSM mode, are shown in the figures below. (Although the global marks R and S are specified, they are not used in the PSM mode.)



● Description of the command

The GLMPOS command specifies execution of overlay writing in the DW mode as well as in the PSM mode by using global marks on the substrate for position alignment, and specifies the positions of global marks P, Q, R, and S on the substrate. The global marks are located at the specified positions on the substrate to correct such problems as positional shift, expansion or shrinkage, and rotation of the substrate. In the DW mode,

the global marks P and any one of Q, R and S, that is, two marks in total, must be specified.

 Global marks P and Q must be given for the models JBX-9000MV, JBX-9300FS and JBX-5500 series.

In the PSM mode, the global marks P and Q must be specified. The coordinate system in which the center of the substrate is the origin is used for positional specification. The X and Y coordinate values increase toward the right and top, respectively.

- ① px
X coordinate of the global mark P position.
- ② py
Y coordinate of the global mark P position.
- ③ qx
X coordinate of the global mark Q position.
- ④ qy
Y coordinate of the global mark Q position.
- ⑤ rx
X coordinate of the global mark R position.
- ⑥ ry
Y coordinate of the global mark R position.
- ⑦ sx
X coordinate of the global mark S position.
- ⑧ sy
Y coordinate of the global mark S position.

4.2.5 GLMP

(Optional in direct writing and PSM writing)

Format:	
GLMP	w,l[, [t] [,f]]
	①② ③ ④
Example:	
GLMP	3.0,1500.0

● Description of the example

This statement specifies that global mark P has a mark width of 3.0 μm and a mark length of 1500.0 μm .

● Description of the command

The GLMP command specifies the shape of global mark P. If this command is not given, the system will use the values assigned to the same parameters in the CALIB program.

- ① w
Mark width of global mark P.
- ② l
Mark length of global mark P.

③ t

Mark type (basic shape) of global mark P.

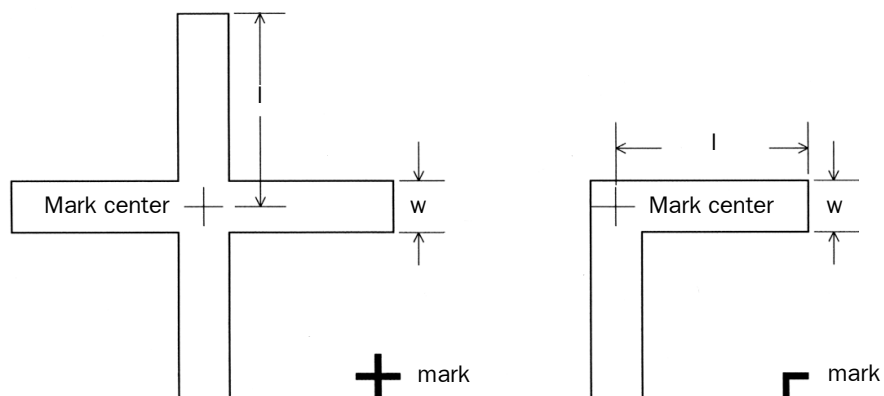
t	Mark type
0	⊕ (default)
1	└

④ f

Orientation of global mark P.

f	Orientation	Type
0	Basic form (default)	⊕ └
1	Rotated 90 degrees CCW	⊕ └
2	Rotated 180 degrees CCW	⊕ └
3	Rotated 270 degrees CCW	⊕ └

 Mark widths and lengths are defined by the mark type as shown below.



4.2.6 GLMQRS

(Optional in direct writing and PSM writing)

Format:	
GLMQRS	w,l[, [t] [,f]]
	①② ③ ④
Example:	
GLMQRS	3.0,1500.0

● Description of the example

Global marks Q, R, and S each has a mark width of 3.0 μm and a mark length of 1500.0 μm .

● Description of the command

The GLMQRS command specifies the shape of global marks Q, R, and S. If this command is not given, the system will use the parameter values specified by the GLMP command.

① w

Mark width of global marks Q, R, S.

② l

Mark length of global marks Q, R, S.

③ t

Type (basic shape) of global marks Q, R, S. If you omit this parameter, the system will use the parameter value specified by the GLMP command.

t	Mark type
0	+
1	┐

④ f

Orientation of global marks Q, R, S. If you omit this parameter, the system will use the parameter value specified by the GLMP command.

f	Orientation	Type
0	Basic form (default)	+ ┐
1	Rotated 90 degrees CCW	+ ┘
2	Rotated 180 degrees CCW	+ └
3	Rotated 270 degrees CCW	+ ┙

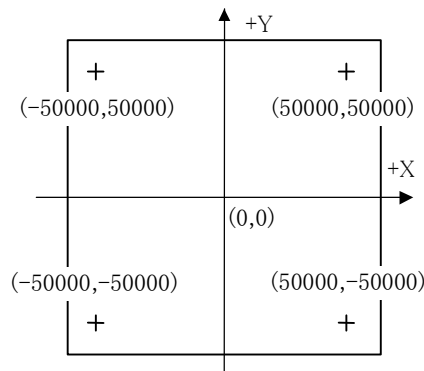
4.2.7 PLMPOS

(Required in phase shift mask pattern writing)

Format:	
PLMPOS	(x1,y1),(x2,y2),(x3,y3)
	① ②
—	[(x4,y4)] [... [(x61,y61)]]
Example:	
PLMPOS	(-50000.0,50000.0),(50000.0,50000.0),
—	(50000.0,-50000.0),(-50000.0,-50000.0)

● Description of the example

This statement specifies the positions on the substrate of plate marks used in the phase shift mask pattern writing (PSM) as shown below.



● Description of the command

The PLMPOS command specifies the positions of plate marks that are located on the substrate and used as alignment marks for overlay writing in the PSM mode. The plate marks are a group of marks located at the specified positions on the substrate and are used to correct for not only such problems as position shift, expansion or shrinkage, rotation, and height change of the substrate, but also, localized dislocation on the substrate plane. The number of plate marks to be specified can be from 3 to 61. Specify the mark positions using the substrate center coordinate system in which the center of the substrate is the origin. The X and Y coordinate values increase toward right and top, respectively.

-  The PLMPOS command is only applicable if the GLMPOS command is specified.
-  The number of plate marks depends on the degree of the approximation for correction. If you use the first-degree approximation, at least three plate marks are required. In the same way, at least six plate marks are required for the second-degree approximation, and at least ten plate marks for the third-degree approximation.

① xn
X coordinate of the plate mark n position.

② yn
Y coordinate of the plate mark n position.

4.2.8 PLMARK

(Optional in phase shift mask pattern writing)

Format:	
PLMARK	w,l ①②
Example:	
PLMARK	3.0,40.0

- **Description of the example**

This statement specifies that the width and the length of the plate mark to be used are 3.0 μm and 40.0 μm , respectively.

- **Description of the command**

The PLMARK command specifies the shape of the plate mark. If this command is not given, the system will use the values assigned to the same parameter in the CALIB program.

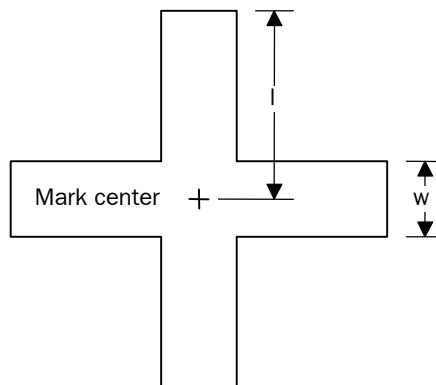
① w

Mark width of the plate mark.

② l

Mark length of the plate mark.

 The definitions of mark width and mark length are shown below.



4.2.9 PATH

(Required)

Format:			
PATH	{	[/P]/[/S]}	[p]
	①	②	③
or			
PATH	{	[/P]/[/S]}	['p']
	①	②	③
Example:			
PATH		JEOL	(1)
PATH	/P	'2JEOL'	(2)
PATH	/P/S	JEOL	(3)

● Descriptions of the examples

- (1) Specifies the start of the writing path named JEOL.
- (2) Specifies not only the start of the writing path 2JEOL, but also the proximity effect correction to be carried out in that writing path.
- (3) Specifies not only the start of the writing path JEOL, but also the proximity effect correction and the field sorting (smashing) to be carried out in that writing path.

● Description of the command

The PATH command declares the start of pattern writing defined by a writing path and specifies whether or not to execute the proximity effect correction and the field sorting (smashing) in that writing path.

When the jobdeck has more than one writing path definition, the system will start e-beam writing beginning with the first writing path definition in order.

① /P

If your e-beam machine is JBX-9000MV or JBX-3030MV, you can optionally use this command qualifier. It specifies execution of proximity effect correction in the e-beam writing path.

② /S

If your e-beam machine is JBX-9000MV or JBX-3030MV, you can optionally use this command qualifier. It specifies execution of the field sorting (smashing) in the e-beam writing path.

③ p

Path name used to identify the e-beam writing path. If you omit this parameter, no path name is given to the e-beam writing path. If the character string that indicates the path name begins with a character other than a letter, enclose the entire character string in single quotes ('').*

 In the e-beam writing routine, you can specify the execution of calibration (INITIAL CALIB) at the beginning of a writing path, and periodic calibration (CYCLIC CALIB) in the writing path. If you specify the execution of the proximity effect correction, the system will carry out not only the proximity effect correction

* This description is available only for the JBX-3030MV.

of individual chip patterns, but also the mutual proximity effect correction of chip patterns written in the writing path definition. Various parameters that are required for the execution of proximity effect correction are to be specified for each substrate in the schedule file. When the parameters for proximity effect correction are not specified in the schedule file, the system will use their default parameters specific to the e-beam lithography system.

Each chip pattern written in a writing path is written sequentially.

The writing sequence is made in X axis zigzag mode from larger to smaller Y axis values around the center position of the material on which the chip center is located. However, if you specify the field sorting (smashing) to be carried out in the writing path, the chip pattern written in the writing path is written sequentially for each writing field composing the chip pattern. The writing sequence is made in X axis zigzag mode from larger to smaller Y axis values around the center position of the material on which the writing field is written.

- ☞ The proximity-effect correction writing is an optional feature for the JBX-9000MV and JBX-3030MV. For information about proximity-effect correction, refer to the “GENERAL STATEMENT,” or the manual, “LITHOGRAPHY OPERATIONS” (for the JBX-9000MV) or “OPERATION” (for the JBX-3030MV) for your lithography system.
- ☞ For information about field sorting (or smashing), refer to the “GENERAL STATEMENT” or “LITHOGRAPHY OPERATIONS” for the lithography system.

4.2.10 SCALE

(Optional)

Format:	
SCALE	sx[,sy] ① ②
Example:	
SCALE	0.99

● Description of the example

A scaling value of 0.99 will be applied to all chips designated in a writing path definition in fine pitch control writing.

- ✂ For information about the fine pitch control writing, refer to the “GENERAL STATEMENT” (for models except JBX-3030MV) or “MAINTENANCE” (for the JBX-3030MV) or “LITHOGRAPHY OPERATIONS” (for models except JBX-3030MV) or “OPERATION” (for the JBX-3030MV) for the lithography system.

● Description of the command

The SCALE command executes “fine pitch control writing”, “minimum increment variable writing” and specifies the magnification or reduction ratio for the writing. The system magnifies or reduces chips with reference to the chip center but keeps the center position unchanged. When the scaling value specified is 1.0, the system will not execute

the fine pitch control writing. If this command is not given, the system will use the value specified in the SCALE command in jobdeck layer blocks. The SCALE command specification in the common block will override that in jobdeck layer blocks for each writing path definition.

☞ For details about the minimum increment variable writing, see the separate instruction manual.

① **sx**

Scaling ratio in the X direction.

② **sy**

Scaling ratio in the Y direction; if you omit this parameter, the system will use the value of sx above.

● **Supplemental remarks**

The fine pitch control writing function and the minimum increment variable writing function are optional functions. If you did not purchase these functions, you cannot define the SCALE command.

✂ For information about the minimum increment variable writing, refer to the “GENERAL STATEMENT” (for models except JBX-3030MV) or “MAINTENANCE” (for the JBX-3030MV) or “LITHOGRAPHY OPERATIONS” (for models except JBX-3030MV) or “OPERATION” (for the JBX-3030MV) for the lithography system.

4.2.11 ARRAY

(Optional)

Format:

[a:] ARRAY (x,m,p)[,...]/(y,n,q)[,...]

①

②③④

⑤⑥⑦

Example:

ARRAY (0,5,10000)/(0,5,10000)

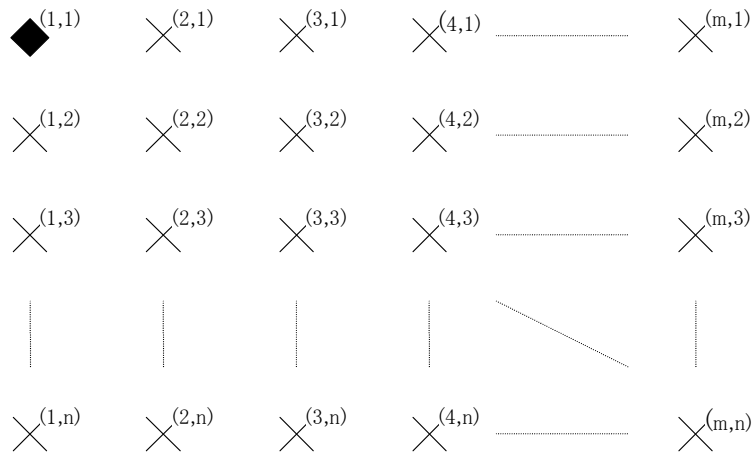
● **Description of the example**

An array composed of 5×5 chips is arranged starting at the substrate center (0,0), 5 chips in the rightward (X) direction with a pitch of 10000 μm and 5 chips in the downward (Y) direction with a pitch of 10000 μm .

● **Description of the command**

The ARRAY command declares the start of a chip array definition, and specifies an arrangement of positions for chip writing.

Coordinates are assigned to array points, starting at the top left ◆ (1, 1) as shown below.



- ① a
Array number; it is referred to when an array number at a lower level is assigned to one at a higher level of nesting for a chip array by the ASSIGN command. This parameter can be omitted when this chip array definition is not referred to by any other chip array definition.
 For details about the nesting of chip array definitions, see the ASSIGN command.
- ② x
X coordinate of array start point with reference to the coordinate system having its origin at the substrate center or the start point of arrays on a higher level of nesting.
- ③ m
Number of array points along the X direction.
- ④ p
Pitch between adjacent array points along the X direction.
- ⑤ y
Y coordinate of array start point with reference to the coordinate system having its origin at the substrate center or the start point of arrays on a higher level of nesting.
- ⑥ n
Number of array points along the Y direction.
- ⑦ q
Pitch between adjacent array points along the Y direction.

4.2.12 CHMPOS

(Optional in direct writing)

Format:	
CHMPOS	M1=(x1,y1) [,M2=(x2,y2)
	① ②
–	,M3=(x3,y3) [,M4=(x4,y4)]]
Examples:	
CHMPOS	M1=(–10000,10000)
–	,M2=(–9000,10000),M3=(–10000,9000) (1)
CHMPOS	M1=(–10000,10000),M2=(10000,10000)
–	,M3=(10000,–10000),M4=(–10000,–10000) (2)

● Descriptions of the examples

- (1) Three chip marks M1, M2, and M3 are located at positions (–10000 μm , 10000 μm), (–9000 μm , 10000 μm), and (–10000 μm , 9000 μm), respectively, referring to the origin that coincides with each array point in the chip array defined by the ARRAY command immediately before.
- (2) In the same way, four chip marks M1, M2, M3, and M4 are located at positions (–10000 μm , 10000 μm), (10000 μm , 10000 μm), (10000 μm , –10000 μm), and (–10000 μm , –10000 μm), respectively.

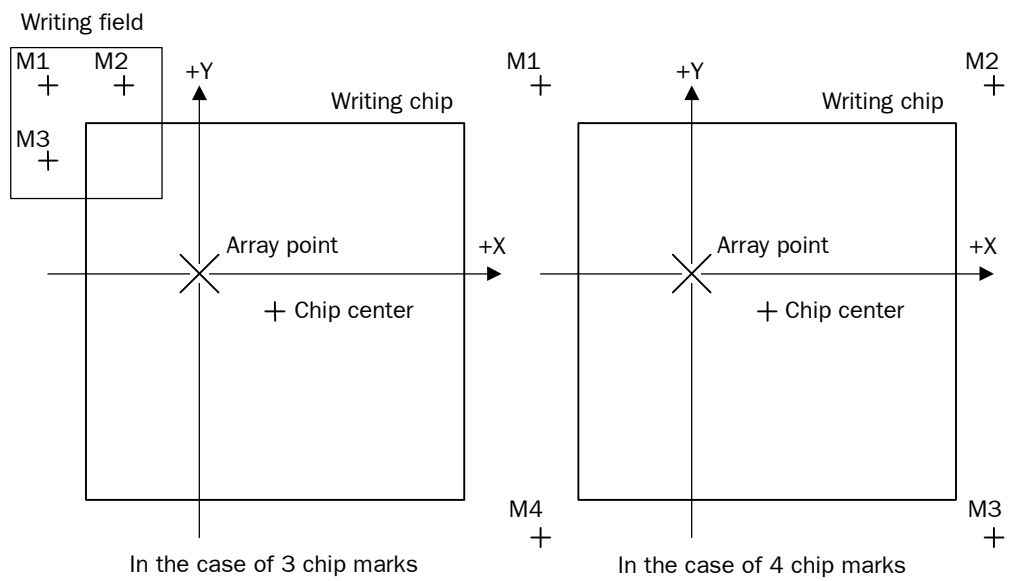
● Description of the command

The CHMPOS command specifies the positions of the chip marks M1, M2, M3, and M4 on the substrate that are used in the direct writing mode.

Chip marks are a group of alignment marks located at the specified positions on the substrate to correct the deflection field (writing field) by adjusting the shift, gain, and rotation of the deflector of the e-beam lithography system. Positions are specified by the coordinate system having its origin at each array point of the chip array defined by the ARRAY command issued most recently. In the array point center coordinate system, the X and Y coordinate values increase toward the right and the top, respectively.

If this command is not given, the system will not execute chip mark detection; consequently, various types of writing corrections using detected results will not be made.

 In the JBX-6000FS, the JBX-6300 series and the JBX-5500 series, the positions of one chip mark (M1) or three chip marks (M1 to M3) are specified. For the latter case, all three chip marks have to be positioned in such a way that they are included in one writing field for a lithography system. In the JBX-6300 series supporting the 4 chip marks alignment program, as well as in the JBX-9000MV and JBX-9300FS, the positions of one chip mark (M1) or four chip marks (M1 to M4) have to be specified. Please note that, for the JBX-9000MV and JBX-6300 series, please purchase the 4 chip marks alignment program.



- ① x_n
X coordinate of the chip mark M_n position.
- ② y_n
Y coordinate of the chip mark M_n position.

4.2.13 CHMARK

(Optional in direct writing)

Format:	
CHMARK	w,l[, [t] [,f]]
	① ② ③ ④
Example:	
CHMARK	3.0,40.0

- **Description of the example**

The width and the length of the chip marks are 3.0 μm and 40.0 μm , respectively.

- **Description of the command**

The CHMARK command specifies the shape of the chip marks. If this command is not given, the system will use the values specified for the same parameter in the calibration program.

① w

Mark width of chip mark.

② l

Mark length of chip mark.

③ t

Type (basic shape) of chip mark.

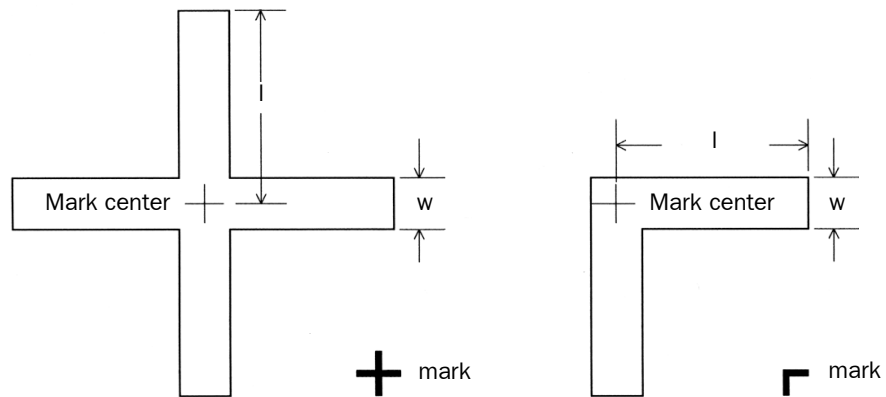
t	Mark type
0	⊕ (default)
1	┐

④ f

Orientation of chip mark.

f	Orientation	Type
0	Basic form (default)	⊕ ┐
1	Rotated 90 degrees CCW	⊕ ┐
2	Rotated 180 degrees CCW	⊕ ┐
3	Rotated 270 degrees CCW	⊕ ┐

 The mark widths and mark lengths of chip marks are defined as shown below.



4.2.14 ASSIGN

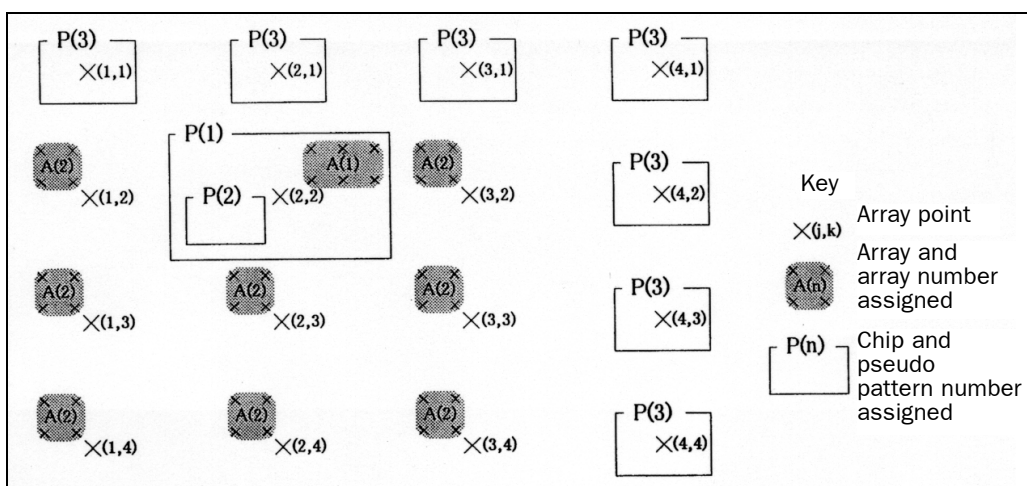
(Optional)

Format:		
ASSIGN	$\{P(i) \mid A(a)\} [+...] \rightarrow$	
–	$[(\textcircled{1})[(\textcircled{4})(j[-j'],k[-k'])],\dots][t,t1][\textcircled{6}][\textcircled{7}][\dots]]$	
ASSIGN	$\{P(i) \mid A(a)\} [+...] \rightarrow$	
–	$[(\textcircled{1})[(\textcircled{4})(j[-j'],k[-k'])],\dots][t,t']][\dots]]$	
Example:		
ASSIGN	$P(3) \rightarrow ((*,1-2),(3-4,*),MDLT01)$	(1)
ASSIGN	$A(2) \rightarrow (1-3,2-4)$	(2)
ASSIGN	$P(1)+P(2)+A(1) \rightarrow ((2,2),'3030MV')$	(3)

● Description of the example

- (1) The chip of pseudo pattern number 3 is assigned, together with the shot time modulation table having table name MDLT01, to all array points in the X direction of the array points 1 and 2 in the Y direction, and to the array points 3 and 4 in the X direction of all array points in the Y direction defined by a chip array definition.
- (2) In addition, the array of array number 2 is assigned to the array points 1 through 3 in the X direction of the array points 2 through 4 in the Y direction in place of pseudo pattern number 3. The assignments in (1) to those array points are invalidated.
- (3) Furthermore, for array point (2, 2), the previous assignment is invalid; the chips of pseudo pattern numbers 1 and 2, and the chip array definition for array number 1, are assigned together with the shot time modulation table (table name: 3030MV).

See the figure below.



● Description of the command

The ASSIGN command assigns one or more chips to arbitrary array points defined by chip array definitions. It also can be used to assign other multiple chip array definitions to arbitrary array points defined by chip array definitions, and that is called nesting in chip array definitions.

① i

Pseudo pattern number of the chip to be assigned.

② a

Array number of the chip array definition to be assigned.

③ →

Symbol that indicates that the pseudo pattern numbers or the array numbers listed on the left side of this symbol are to be assigned to the array points listed on the right side of the symbol.

④ j

Array points in the X direction to which chips and chip array definitions are to be assigned; $j - j'$ ($j < j'$) designates array points from a certain point (j) to another (j'); an asterisk (*) means assignment of all array points in the row.

⑤ k

Array points in the Y direction to which chips and chip array definitions are to be assigned; $k - k'$ ($k < k'$) designates array points from a certain point (k) to another (k'); an asterisk (*) means assignment of all array points in the column.

⑥ t

Name of shot time modulation table to be applied to one or more array points assigned. The combination of array points and shot time modulation table name is enclosed in parentheses: $((j_1, k_1), (j_2, k_2), \dots, t)$. If the character string that indicates the shot time modulation table name begins with a character other than a letter, enclose the entire character string in single quotes (')*.

⑦ t1 (valid only for the JBX-6300 series)

Name of the single line shot time modulation table to be applied to one or more array points are assigned. If this assignment is omitted, shot time modulation is

* This description is available only for the JBX-3030MV.

performed in the same way as rectangular shot time modulation. The combination of array points, and the names of rectangular shot time modulation table and of single line shot time modulation table is enclosed in parentheses ((j1,k1), (j2,k2),...,t,tl).

■ Supplemental remarks

● Description example in the case of the JBX-6300 series

Example 1: If rectangular data and single line data coexist in a writing pattern and respective shot rank are the same:

```

ARRAY      (-1000,3,500)/(1000,3,500)
ASSIGN      P(1)    ->      ((1,1),(1,2),SHOT01,SHOT02)
:
P(1)  'TEST'
RESIST      100,1000,A                      ;unit=μC/cm^2, μC/cm^2
SHOT01:     MODULAT((0,-10))
SHOT02:MODULAT((0,100))

```

Example 2: If rectangular data and single line data coexist in a writing pattern and respective shot rank are different:

```

ARRAY      (-1000,3,500)/(1000,3,500)
ASSIGN      P(1)    ->      ((1,1),(1,2),SHOT01,SHOT02)
:
P(1)  'TEST'
RESIST      100,1000,A                      ;unit=μC/cm^2, μC/cm^2
SHOT01:     MODULAT((0,-10))
SHOT02:     MODULAT((0,100))

```

Example 3: If rectangular data and single line data coexist in a writing pattern and respective shot rank are omitted:

```

ARRAY      (-1000,3,500)/(1000,3,500)
ASSIGN      P(1)    ->      ((1,1),(1,2),SHOT01,SHOT01)
:
P(1)  'TEST'
RESIST      100,1000,A                      ;unit=μC/cm^2, μC/cm^2
SHOT01:     MODULAT((0,-10))

```

The line sensitivity of the resist is ignored and the shot rank table for rectangle is applied.

Example 4: If a writing pattern contains only rectangular data:

```

ARRAY      (-1000,3,500)/(1000,3,500)
ASSIGN      P(1)    ->      ((1,1),(1,2),SHOT01)
:

```

```

P(1)  'TEST'
RESIST                100,1000,A      ;unit= $\mu\text{C}/\text{cm}^2$ ,  $\mu\text{C}/\text{cm}^2$ 
SHOT01:  MODULAT((0,-10))

```

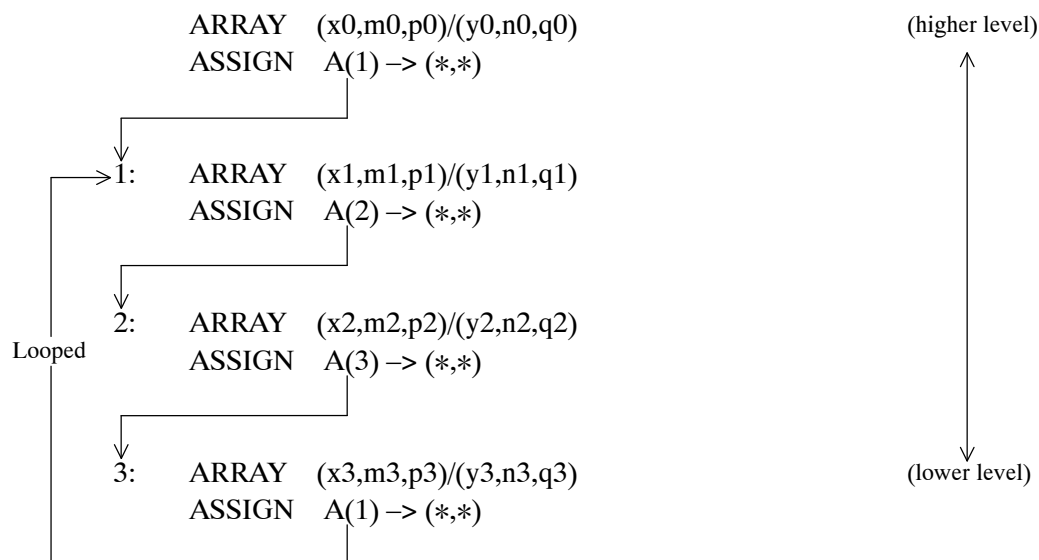
 However, you cannot omit the assignment of the line sensitivity of the resist.

- When more than one array point (j, k), or more than one combination of array points and a shot time modulation table name ((j₁, k₁), (j₂, k₂), ..., t) are to be assigned in one ASSIGN command, all of them are enclosed in parentheses ((j, k), ((j₁, k₁), (j₂, k₂), ..., t), ...).
- When more than one ASSIGN command is specified at the same array point, chips and chip array definitions assigned by the succeeding ASSIGN commands will override the previous ones.
- Up to 10 levels of nesting can be defined for a chip array definition.

	ARRAY	(x0,m0,p0)/(y0,n0,q0)		(higher level)
	ASSIGN	A(1) -> (*,*)	level 1	↑ ↓
1:	ARRAY	(x1,m1,p1)/(y1,n1,q1)		
	ASSIGN	A(2) -> (*,*)	level 2	
2:	ARRAY	(x2,m2,p2)/(y2,n2,q2)		
	ASSIGN	A(3) -> (*,*)	level 3	
:				
:				
9:	ARRAY	(x9,m9,p9)/(y9,n9,q9)		
	ASSIGN	A(10) -> (*,*)	level 10	
10:	ARRAY	(x10,m10,p10)/(y10,n10,q10)		
	ASSIGN	P(1) -> (*,*)		

 The chip array definition assigned to level 10 is useless (this would exceed the level number).

You cannot make a job schedule such that a chip array definition at a lower level assigns to a chip array definition at a higher level; that is, the assignment in which nesting structures make a loop cannot be accepted.



 This assignment that makes a loop cannot be made.

Also, chip array definitions in other writing path definitions cannot be assigned.
The shot time modulation table will be in effect for all chips written within chips and chip array definitions assigned.

4.2.15 SKIP

(Optional)

Format:		
SKIP	$[(\textcircled{1}j[-j'],k[-k'])[\dots]]$	
		(1)
Example:		(2)
ASSIGN	P(3) → ((*,1-2),(3-4,*),MDLT01)	(3)
SKIP	(1-3,2-4)	
ASSIGN	P(1)+P(2)+A(1) → (2,2)	

● Description of the example

- (1) This ASSIGN statement assigns the chip of pseudo pattern number 3, together with the shot time modulation table having table name MDLT01, to all array points in the X direction of the array points 1 and 2 in the Y direction, and to the array points 3 and 4 in the X direction of all array points in the Y direction defined by a chip array definition.
- (2) This SKIP statement cancels the pattern assignment that was made to the array points 1 to 3 in the X direction of the array points 2 to 4 in the Y direction from the array points defined by the pseudo pattern number 3.
- (3) This ASSIGN statement assigns the chips of the pseudo pattern numbers 1 and 2, and of the array number 1 to the array point (2, 2) in place of the previous assignment.

● Description of the command

The SKIP command cancels the assignment of the chips of arbitrary array points in a chip array definition and the assignment of chip array definitions.

① j

Array points in the X direction of which chips and chip array definitions are to be canceled; j – j' (j < j') designates array points from a certain point (j) to another (j'); an asterisk (*) means all array points in the row.

② k

Array points in the Y direction of which chips and chip array definitions are to be canceled; k – k' (k < k') designates array points from a certain point (k) to another (k'); an asterisk (*) means all array points in the column.

 When ASSIGN and SKIP commands refer to the same array point, whichever command is given last will be in effect. The SKIP command cancels assignment of the shot time modulation table for the specified array points as well.

4.2.16 AEND

(Optional)

Format: AEND (no parameters)

- **Description of the command**

The AEND command ends the chip array definition that began with the ARRAY command.

4.2.17 CHAR

(Optional)

Format: CHAR/i (x,y){[, [a] [,b]] [, [m] [,r]]} ① ②③ ④ ⑤ ⑥ ⑦
--

Example: CHAR/1 (-50000.0,50000.0)

- **Description of the example**

The character string defined by the character string number 1 will be written at the position $X = -50000.0 \mu\text{m}$ and $Y = 50000.0 \mu\text{m}$ in the substrate center coordinate system. Normal characters in horizontal arrangement are employed as defaults.

- **Description of the command**

The CHAR command specifies the writing position and attributes of a character string. For the position of the character string, specify the center position of the character string in the coordinate system whose origin is the center of the substrate (the substrate center coordinate system; X and Y values increase toward the right and top, respectively). For the attributes of the character string, you can specify numeric values and/or letters to indicate the direction of the character string. The method of specifying the attributes of the character string differs depending upon the value you specified for the parameter sf of the OPTION command.

① i

Character string number.

i	Description
1 – 4	User defined character strings
T	Date and time
M	Model name
D	Character string specified by the DEVINF command for designating a device
L	Character string specified by the LAYER command for designating a layer

② x

X coordinate of the position where the character string is to be written.

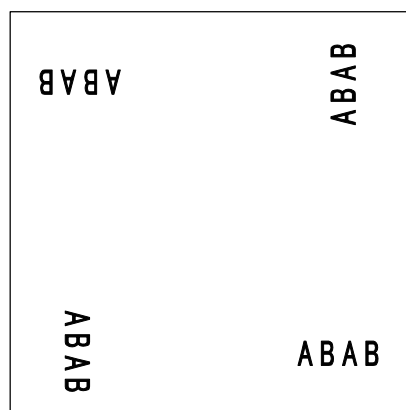
③ y

Y coordinate of the position where the character string is to be written.

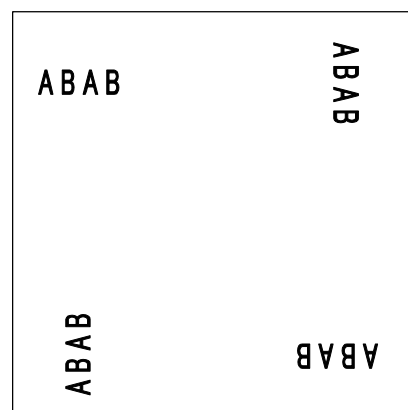
④ a

Specifies an attribute of the character string (normal or inverted), if the parameter sf of the OPTION command is 0.

a	Attribute
0	Normal (default)
1	Inverted



Example of normal characters

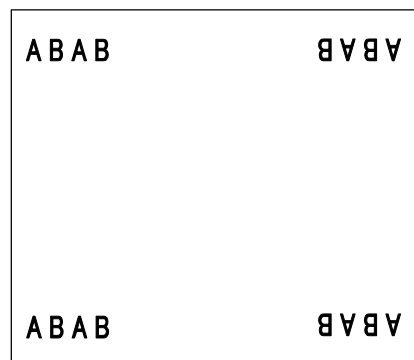


Example of inverted characters

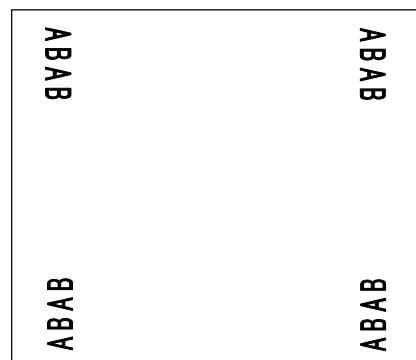
⑤ b

Specifies an attribute of the character string (horizontal or vertical), if the parameter sf of the OPTION command is 0.

b	Attribute
0	Horizontal (default)
1	Vertical



Example of horizontal arrangement

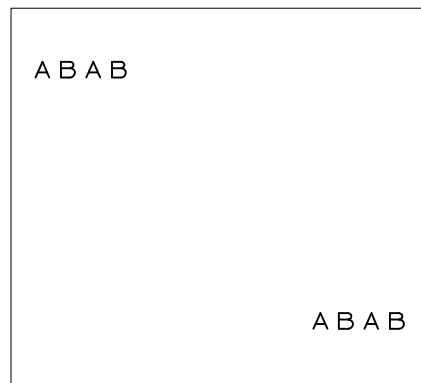


Example of vertical arrangement

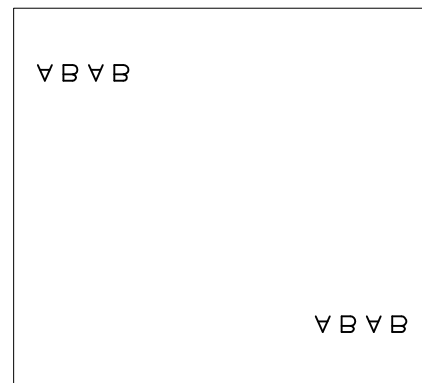
⑥ m

Specifies attributes of the character string (whether or not to perform mirror inversion, and an inversion axis), if the parameter sf of the OPTION command is 1.

m	Attributes of the character string
0	Normal character string (default)
X	Mirrors the character string with respect to a horizontal axis that passes through the center of the character string.
Y	Mirrors the character string with respect to a vertical axis that passes through the center of the character string.



Example of normal characters
(without mirror inversion)



Example of X-axis inverted characters



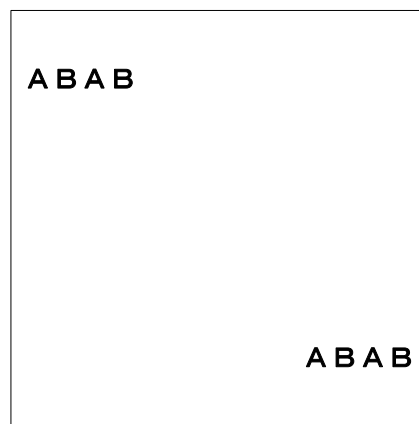
Example of Y-axis inverted characters

This parameter is used prior to the specification of the parameter r.

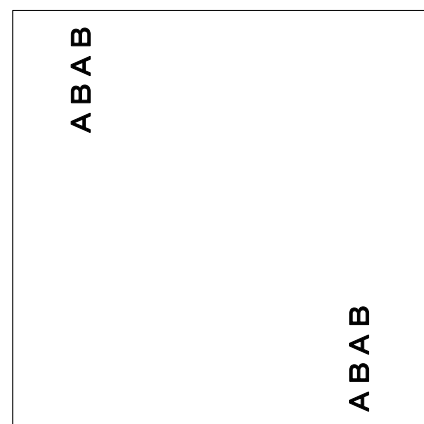
⑦ r

Specifies attributes of the character string (whether or not to perform rotation, and a rotation angle), if the parameter sf of the OPTION command is 1.

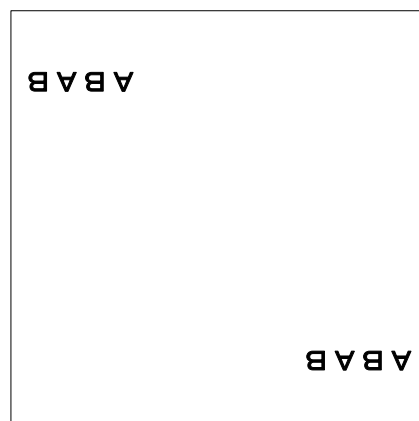
r	Attributes of the character string
0	Normal character string (default)
90	Rotates the character string 90 degrees counterclockwise around the center of the character string.
180	Rotates the character string 180 degrees counterclockwise around the center of the character string.
270	Rotates the character string 270 degrees counterclockwise around the center of the character string.



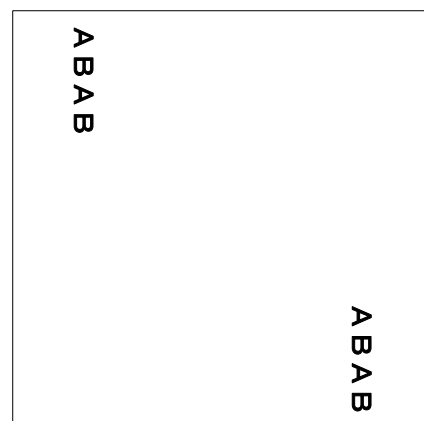
Example of normal characters
(character strings rotated 0 degrees)



Example of 90-degree rotated
characters



Example of 180-degree rotated
characters



Example of 270-degree rotated
characters

This parameter is used posterior to the specification of the parameter m.

4.2.18 PEND

(Required)

Format: PEND (no parameters)
--

● Description of the command

The PEND command ends the writing path definition that began with the PATH command.

4.2.19 P

(Optional)

Format:
P(i) 'p' [(δx , δy)] [,m,n[,m',n']]
① ② ③ ④ ⑤ ⑥
— [,CS=(csx,csy)]
⑦ ⑧

Example:

P(1)	'jeol.j5x'(-100,100)	(1)
P(2)	'JEOL'(-100,100)	(2)
P(3)	'jeol.j5x',MULTI,3	(3)
P(4)	'jeol.j5x',GATHER,3,OVERLAP,2	(4)

● Descriptions of the examples

- (1) The chip pattern stored in the file jeol.j5x will be written at the position shifted ($-100 \mu\text{m}$, $100 \mu\text{m}$) from the array points to which the pseudo pattern number 1 is assigned. The chip center for writing will be aligned at this new position.
 The shift amounts increase rightward and upward along the X and Y axes, respectively.
- (2) The chip pattern having the ID name JEOL will be written at the position shifted ($-100 \mu\text{m}$, $100 \mu\text{m}$) from the array points to which the pseudo pattern number 2 is assigned. The chip center for writing will be aligned at this new position.
- (3) The chip pattern stored in the file jeol.j5x will be written using the triple field shift overlap writing mode, with aligning the chip center at the array points to which the pseudo pattern number 3 is assigned.
- (4) Not only will the chip pattern stored in the file jeol.j5x be written using the field combine writing mode that writes 3×3 fields as a block but also each field will be written using the double overlap writing mode, with aligning the chip center at the array points to which the pseudo pattern number 4 is assigned.

● Description of the command

The P command writes a chip pattern at the array points to which the pseudo pattern number is assigned. It also specifies the shift amounts from the array points, and special writing modes. The P command specified in a jobdeck layer block will override the P command concerning the same pseudo pattern number specified in the common block.

① i

Pseudo pattern number for chip writing.

② p

ID name of the chip pattern data to be written or the name of the file in which chip pattern data to be written are stored.

 For lithography systems controlled by a DEC (now an HP) VAX series computer system, such as the JBX-6000FS, this parameter specifies the pattern data ID name of the chip. For lithography systems controlled by a Compaq (now an HP) AlphaStation computer system, such as the JBX-9000MV, JBX-9300FS, JBX-3030MV, and JBX-6000FS/E, or by a Sun Solaris series computer system, such as the JBX-6300 series and the JBX-5500 series, this parameter specifies the name of the file that contains the chip pattern data.

③ δx

Shift amount, along the X axis, of the writing position to which the chip center is shifted with reference to the array point.

④ δy

Shift amount, along the Y axis, of the writing position to which the chip center is shifted with reference to the array point.

⑤ m

Specifies special writing modes.

m	Special writing mode
MULTI	Specifies execution of the field shift overlap writing mode. In this mode, overlap pattern writing is performed n times, applying one nth of the normal e-beam dose, by changing the writing position on the writing fields of pattern data.
MULTI/FP	Carries out the field shift overlap writing for the entire chip, in order from the first overlap, without mixing the writing fields of different overlaps.
GATHER	Specifies execution of the field combine writing mode. In this mode, the patterns of up to $n \times n$ fields are handled as a single field block and written for pattern data of which the field size is one nth of the maximum field size.
OVERLAP	Specifies execution of the overlap writing mode. In this mode, overlap pattern writing is performed n times, applying one nth of the normal e-beam dose on the writing fields of pattern data.

You can specify GATHER and OVERLAP together. You can also specify MULTI and GATHER together. For such cases, you should apply one to m, and the other to m'. The field combine writing is called the full pass writing. You can specify the full pass writing only in the JBX-3050MV and later models.

 For information about field shift overlap writing, refer to the “GENERAL STATEMENT” or “LITHOGRAPHY OPERATIONS” for each lithography system other than JBX-6000FS/E

 For information about field combine writing, refer to the “GENERAL

STATEMENT” or “LITHOGRAPHY OPERATIONS” for each lithography system other than JBX-6000FS/E.

☞ For information about overlap writing, refer to the “GENERAL STATEMENT” or “LITHOGRAPHY OPERATIONS” for each lithography system other than JBX-6000FS/E.

⑥ n

Specifies the number of overlaps in the overlap writing mode or the field shift overlap writing mode, or specifies the number of fields to be combined along either the X or Y axis in the field combine writing mode. Also, n' is used to specify in the same way as n.

⑦ csx

Specifies the chip size X for the pattern data to write. If there is no pattern data to write, this parameter serves as the default value of the chip size X.

⑧ csy

Specifies the chip size Y for the pattern data to write. If there is no pattern data to write, this parameter serves as the default value of the chip size Y.

✗ When you omit (δx , δy), the chip center for writing will be at array points. When you omit the parameters m and n, special writing modes will not be performed.

✗ Prior to the chip sizes you specified using CS = (csx, csy), the values specified in the pattern data are used.

4.2.20 SPPRM

(Optional; inapplicable in JBX-6000FS.)

Format:	
SPPRM	[m1], [m2], [w], [p] [, [sc] [, [ss] [, [sfs]]]
	① ② ③ ④ ⑤ ⑥ ⑦
Example:	
P(1)	'jeol.j5x', MULTI, 3 (1)
SPPRM	2.0, 4.0, 0.24, 0.12, 2, 2 (2)

● Description of the example

- (1) The preceding P command specifies the file jeol.j5x and specifies to write the chip pattern contained in that file using the triple field shift overlap writing
- (2) For the triple field shift overlap writing, the SPPRM command sets the following conditions: maximum beam size = 2.0 μm , maximum beam area = 4.0 μm^2 , trapezoid writing slit width = 0.24 μm , trapezoid writing pitch = 0.12 μm , writing magnification = 1.0, shot-shift amount for the field shift overlap writing = half of the beam size, subfield-shift amount for the subfield-shift writing = half of the subfield size.

● Description of the command

The SPPRM command specifies shot partition parameters for each writing chip pattern data. The SPPRM command specified in a jobdeck layer block will override the SPPRM command specified in the common block if the SPPRM command in the jobdeck layer

block concerns the same pseudo pattern number.

For the pattern data corresponding to a pseudo pattern number, if the SPPRM command is specified neither in the jobdeck common block nor in a jobdeck layer block, the default parameter values that are provided by the system will be used.

 In the case of the variable shaped e-beam lithography system, shot partition parameters will be performed as beam shaping parameters. In the case of the spot e-beam lithography system, they will be performed as parameters related to the beam scanning range that can be written by a single positioning deflection.

① m1

For the variable shaped e-beam machine, this parameter is used to specify the maximum value of the longer side of the shaped rectangle beam for writing a chip.

For the spot e-beam machine, this parameter is used to specify the maximum value of the side of the square beam scanning area that can be written by a single positioning deflection.

② m2

For the variable shaped e-beam machine, this parameter is used to specify the maximum value of the shaped rectangle beam area for writing a chip.

In the case of the spot e-beam machine, this parameter is automatically set to the square of m1.

 In the case of the spot e-beam lithography system, the specification of the parameter m2 will not be used, even if it is specified.

③ w

For the variable shaped e-beam machine, this parameter is used to specify the length of the shorter side of the shaped rectangle beam for writing a trapezoid pattern.

For the spot e-beam machine, specification of this parameter is inapplicable.

④ p

For the variable shaped e-beam machine, this parameter is used to specify the shift pitch of the shaped rectangle beam in the shorter side direction for writing a trapezoid pattern.

For the spot e-beam machine, specification of this parameter is inapplicable.

⑤ sc

Specifies the magnification or reduction ratio of scaling for pattern writing.

If you omit sc, the magnification or reduction ratios of scaling in the directions of both X and Y axes are set to 1.0.

For the JBX-3040MV and later models, the product of the writing enlargement/reduction ratio specified in the JEOL52 (V3.1) format data and the value of this parameter is the scaling enlargement/reduction ratio to use for writing.

⑥ ss

Specifies the shift amount for partition of the shaped rectangle beam or that for the spot beam scanning area when shot shift writing is performed. The value will be the inverse of the value obtained from the maximum beam area (m2) that corresponds to the pattern shape while each size of the maximum beam area is set to be 1 both in the horizontal direction and in the vertical direction. If you omit this parameter, shot shift writing will not be performed.

If neither overlap writing nor field shift overlap writing is applied to the pattern data, shot shift writing will not also be applied to it.

⑦ sfs

The parameter sfs specifies the subfield-shift amount for the subfield-shift writing. For the parameter value, specify the reciprocal of a fraction whose numerator is 1, assuming that the subfield size is 1. The subfield-shift writing is disabled by default. For pattern data for which the overlap writing and the field shift overlap writing are disabled, the subfield-shift writing is also disabled.

-  This parameter is closely related to writing accuracy, so be very careful when setting it.
-  For information about the shot-shift writing, see the manual, “General Statement” or the manual, “Operation” for your lithography system.
-  The subfield-shift writing is an optional feature for the JBX-3030MV and is a standard feature for the JBX-3040MV and later models. For information about subfield-shift writing, refer to the “GENERAL STATEMENT” or the manual, “OPERATION” for your lithography system.

4.2.21 STGPRM

(Optional/available only in the JBX-3050MV and later models)

Format:

STGPRM [a1] [, [a2] [, [a3] [, [a4], a5]]]

① ② ③ ④ ⑤

Example:

P(1)	'jeol.j5x',MULTI/FP,2	(1)
STGPRM	XRASTL,XRASTL	(2)

● Description of the example

- (1) The preceding P command specifies a double field shift overlap writing in which the full pass writing applies to the chip pattern contained in the jeol.j5x file.
- (2) In that writing, the substrate stage moves in the X raster sequence from the top left of the pattern to write the pattern of the first overlap, and the substrate stage moves in the X raster sequence from the top left of the pattern to write the pattern of the second overlap.

● Description of the command

The STGPRM command specifies the method of moving the substrate stage in the full pass writing. If the preceding P command specifies the full pass writing, the STGPRM command specifies how to move the substrate stage to write the pattern in the normal writing and the nth writing in the field shift overlap writing. If the full pass writing is not specified, only the a1 parameter takes effect; the writing is performed, with no distinction between the first and second overlaps, using the specified movement method (with XZIGTL specified, the writing is performed in the usual manner).

If the P command specifies the full pass writing and if the STGPRM command is omitted, the default values specified as system parameters are used. If the P command does not specify the full pass writing and if the STGPRM command is omitted, the writing is performed in the usual manner.

① a1

Here you can specify the direction in which to move the stage for the first overlap. You can use the following parameters to determine the movement method.

Parameter	Description
XRASTL	X raster writing from the top left of the pattern
XRASTR	X raster writing from the top right of the pattern
XRASBL	X raster writing from the bottom left of the pattern
XRASBR	X raster writing from the bottom right of the pattern
XZIGTL	X zigzag writing from the top left of the pattern
XZIGTR	X zigzag writing from the top right of the pattern
XZIGBL	X zigzag writing from the bottom left of the pattern
XZIGBR	X zigzag writing from the bottom right of the pattern

You can omit this parameter; in that case, the default value specified as a system parameter is used.

② a2

Here you can specify the direction in which to move the stage for the second overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is neglected.

③ a3

Here you can specify the direction in which to move the stage for the third overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is neglected.

④ a4

Here you can specify the direction in which to move the stage for the fourth overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is neglected.

⑤ a5

Here you can specify the direction in which to move the stage for the fifth overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is neglected.

● **Supplementary notes**

● Omitting parameters

With both /FP and STGPRM omitted

⇒The conventional writing is performed.

With /FP omitted and STGPRM specified

⇒Only the first parameter of STGPRM takes effect, and using the movement method of that parameter, the writing is performed with no distinction between the first and second overlaps.

With /FP specified and STGPRM omitted

⇒The writing is performed using the default STGPRM value specified as a system parameter.

- Handling the SMASH writing

For a path for which SMASH is specified, the parameter that determines the movement method must be the same for all the patterns in the path. Any specification that is different will cause an error in the data-preparation stage before writing.

4.2.22 C

(Optional)

Format:	
C(i)	's'
①	②
Example:	
C(1)	'JEOL'

- **Description of the example**

The character string “JEOL” will be written on the substrate at the position and with attributes designated by character string number 1 of the CHAR command (CHAR/1).

- **Description of the command**

The C command specifies not only the position and attributes of a character string to be written on the substrate by designating a character string number of the C command, but also the character string to be written by giving it. The C command specified in a jobdeck layer block will override the C command specified in the common block if the C command in the layer block concerns the same character string number.

① i

Character string number that specifies the position and attributes for the character string.

② s

Character string to be written on the substrate.

4.3 Layer Block Commands

4.3.1 LAYER

(Required)

Format:			
LAYER	/P	n	['s']
	①	②	③
Example:			
LAYER		1	(1)
LAYER/P		2,'SHIFT'	(2)

● Descriptions of the examples

- (1) This statement declares the beginning of the layer block defined by layer number 1.
- (2) This statement not only declares the beginning of a layer block defined by layer number 2, but also specifies the phase shift mask pattern writing (PSM) mode for writing.

● Description of the command

The LAYER command declares the beginning of a layer block, and specifies whether or not to use the PSM mode for writing, the layer number, and the layer name to be written as a character string, by designating parameters.

① /P

Specifies execution of e-beam writing in the PSM mode. This command modifier can optionally be used in the JBX-9000MV and JBX-3030MV. If neither the GLMPOS command nor the PLMPOS command is in the jobdeck common block, this command will be inapplicable.

② n

Layer number.

③ s

Layer name to be written on the substrate as a character string. The position and attributes of the character string are designated by the CHAR command (CHAR/L).

 As shown below, the applicable writing mode for each layer block will be determined by the combination of commands and parameters that are specified in the jobdeck file.

Jobdeck File			Writing mode for each layer
Common block		Layer block	
GLMPOS	PLMPOS	(LAYER)/P	
Specified	Specified	Specified	Phase shift mask pattern writing
		Not specified	Normal mask pattern writing
	Not specified	*	Direct writing

* If the PLMPOS command is not specified, the parameter /P of the LAYER command will be inapplicable.

Jobdeck File			Writing mode for each layer
Common block		Layer block	
GLMPOS	PLMPOS	(LAYER)/P	
Not specified	*	**	Normal mask pattern writing

4.3.2 CIRCLE

(Optional)

Format:	
CIRCLE	d2[a] [,b]
	① ② ③
Example:	
CIRCLE	5

● Description of the example

The circular cut diameter of the chip array is 5 inches. For the chip array's circular cutout attribute (inside or outside of the circle) and the diameter unit (inch or millimeter), the inch is specified by default. The default writing attribute is the inside of the circle.

● Description of the command

The CIRCLE command specifies implementation of a circular cut of the chip array, the diameter of the circle, and the attributes of the circular cut. The CIRCLE command specified in a layer block will override the JOB command in the jobdeck common block.

① d2

Circular cut diameter of chip array. The sign of d2 designates attributes. If this parameter is not specified, circular cutting is not performed.

d2	Attribute
> 0	Writes chips on the periphery of the circle.
< 0	Does not write chips on the periphery of the circle.
= 0	No circular cutting.

② a

Unit used to specify diameter of circular cut.

a	Unit
I	Inch (default)
M	mm

* If the GLMPOS command is not specified, the PLMPOS command will be inapplicable.

** If the PLMPOS command is inapplicable, the parameter /P of the LAYER command will be inapplicable as well.

③ b

Attribute of writing position of chip array.

b	Attribute
I	Inside of circle (default)
O	Outside of circle

4.3.3 CANCEL

(Optional)

Format:	
CANCEL	p ①
Example:	
CANCEL	T
CANCEL	M

- **Description of the example**

The above two CANCEL commands with the character T and with the character M specify cancellation of writing of character strings representing the date and time, and the machine ID name that are specified by the OPTION command in the common block.

- **Description of the command**

The CANCEL command is used to cancel the writing of character strings representing the date and time, and the machine ID name specified by the OPTION command in the common block. It can also be used in specific layer blocks to cancel writing of the character strings representing the device type specified by the DEVINF command.

① p

Character strings whose e-beam writing is to be canceled.

p	Item to be canceled
T	Character string representing date and time
M	Character string representing machine identification name
D	Character string representing a device

 If you want to cancel the writing of multiple character strings, include multiple CANCEL statement lines contiguously.

4.3.4 MIRROR

(Optional)

Format:	
MIRROR	[p] [c]
	① ②
Example:	
MIRROR	X

- **Description of the example**

The chip writing position is mirrored with respect to the X axis.

- **Description of the command**

The MIRROR command specifies the execution of mirroring of the writing position of chips and character strings with respect to the axis you designate. It also specifies the attributes. Mirroring by the OPTION command specified in the jobdeck common block and mirroring by the MIRROR command specified in the specific layer block can both be in effect. In such a case, the order of mirroring is first execution by the MIRROR command, then by the OPTION command.

① **p**

Specifies mirroring of chip writing positions with respect to the X or Y axis only for the specified layers in a device.

P	Description
	No mirroring (default)
X	X mirroring
Y	Y mirroring

② **c**

Specifies mirroring of character string writing positions with respect to the X or Y axis only for the specified layers in a device.

 When writing position is mirrored in mask writing, the attributes of characters (normal or mirrored) will also change.

c	Description
	No mirroring (default)
X	X mirroring
Y	Y mirroring

4.3.5 SCALE

(Optional)

Format:	
SCALE	sx[,sy] ① ②
Example:	
SCALE	0.99

● Description of the example

The scaling value 0.99 will be applied to all chips designated for fine pitch control writing in a layer.

☞ For information about fine pitch control writing, refer to “Fine Pitch Control Program Operation.”

● Description of the command

The SCALE command executes “fine pitch control writing”, “minimum increment variable writing”, and specifies the ratio of magnification or reduction (scaling value) of chips. The magnification or reduction is carried out with reference to the chip center; however, the center position is kept unchanged. If the scaling value 1.0 is specified, the system will not execute “fine pitch control writing” and “minimum increment variable writing”. If this command is not given, the system will use 1.0 as the scaling value. The SCALE command defined for each writing path definition in the jobdeck common block will override the command defined in layer blocks.

① sx
Scaling value in X direction.

② sy
Scaling value in Y direction; if you omit this parameter, the system will use the value of sx above.

✂ For information about the minimum increment variable writing, refer to the “GENERAL STATEMENT” (for models except JBX-3030MV) or “MAINTENANCE” (for the JBX-3030MV) or “LITHOGRAPHY OPERATIONS” (for models except JBX-3030MV) or “OPERATION” (for the JBX-3030MV)for the lithography system.

4.3.6 P

(Optional)

Format:	
P(i)	'p'[(δx , δy)] [,m,n[,m',n']]
①	② ③ ④ ⑤ ⑥
—	[,CS=(csx,csy)]
	⑦ ⑧
Example:	
P(1)	'jeol.j5x'(-100,100) (1)
P(2)	'JEOL'(-100,100) (2)
P(3)	'jeol.j5x',MULTI,3 (3)
P(4)	'jeol.j5x',GATHER,3
—	,OVERLAP,2 (4)

● Descriptions of the examples

- (1) The chip pattern stored in the file jeol.j5x will be written at the position shifted (−100 μm , 100 μm) from the array points to which the pseudo pattern number 1 is assigned. The chip center for writing will be aligned at this new position.
✍ The shift amounts increase rightward and upward along the X and Y axes, respectively.
- (2) The chip pattern having the ID name JEOL will be written at the position shifted (−100 μm , 100 μm) from the array points to which the pseudo pattern number 2 is assigned. The chip center for writing will be aligned at this new position.
- (3) The chip pattern stored in the file jeol.j5x will be written using the triple field shift overlap writing mode, aligning the chip center at the array points to which the pseudo pattern number 3 is assigned.
- (4) The chip pattern stored in the file named jeol.j5x will be not only written using the field combine writing mode that writes 3 × 3 fields as a block but also written at each field using the double overlap writing mode, aligning the chip center at the array points to which the pseudo pattern number 4 is assigned.

● Description of the command

The P command writes a chip pattern at the array points to which the pseudo pattern number is assigned. It also specifies the shift amounts from the array points, and special writing modes. The P command specified in a jobdeck layer block will override the P command specified in the common block if the P command in the layer block concerns about the same pseudo pattern number.

① i

Pseudo pattern number for chip writing.

② p

ID name of the chip pattern data to be written or the name of the file in which chip pattern data to be written are stored.

✍ For lithography systems controlled by a COMPAQ (now an HP) VAX series computer system, such as the JBX-6000FS, this parameter specifies the pattern data ID name of the chip. For lithography systems controlled by a Compaq (now an HP)

AlphaStation computer system, such as the JBX-9000MV, JBX-9300FS, JBX-3030MV, and JBX-6000FS/E, or by a Sun Solaris series computer system, such as the JBX-6300 series and the JBX-5500 series, this parameter specifies the name of the file that contains the chip pattern data.

③ δx

Shift amount along the X axis of the writing position to which the chip center is shifted with reference to the array point.

④ δy

Shift amount along the Y axis of the writing position to which the chip center is shifted with reference to the array point.

⑤ m

Specifies special writing modes.

m	Special writing mode
MULTI	Specifies execution of the field shift overlap writing mode. In this mode, overlap pattern writing is performed n times, applying one nth of the normal e-beam dose, by changing the writing position on the writing fields of pattern data.
MULTI/FP	Carries out the field shift overlap writing for the entire chip, in order from the first overlap, without mixing the writing fields of different overlaps.
GATHER	Specifies execution of the field combine writing mode. In this mode, the patterns of up to $n \times n$ fields are handled as a single field block and written for pattern data of which the field size is one nth of the maximum field size.
OVERLAP	Specifies execution of the overlap writing mode. In this mode, overlap pattern writing is performed n times, applying one nth of the normal e-beam dose on the writing fields of pattern data.

You can specify GATHER and OVERLAP together. You can also specify MULTI and GATHER together. For such cases, you should apply one to m, and the other to m'. The MULTI/FP writing is called the full pass writing. You can specify the full pass writing only in the JBX-3050MV and later models.

☞ For information about field shift overlap writing, refer to the “GENERAL STATEMENT” or “LITHOGRAPHY OPERATIONS” for each lithography system other than JBX-6000FS.

☞ For information about field combine writing, refer to the “GENERAL STATEMENT” or “LITHOGRAPHY OPERATIONS” for each lithography system other than JBX-6000FS.

☞ For information about overlap writing, refer to the “GENERAL STATEMENT” or “LITHOGRAPHY OPERATIONS” for each lithography system other than JBX-6000FS.

⑥ n

Specifies the number of overlaps in the overlap writing mode or the field shift overlap writing mode, or specifies the number of fields to be combined along either the X axis or Y axis in the field combine writing mode. Also, n' is used to specify in the same as n.

⑦ csx

Specifies the chip size X for the pattern data to write. If there is no pattern data to write, this parameter serves as the default value of the chip size X.

⑧ csy

Specifies the chip size Y for the pattern data to write. If there is no pattern data to write, this parameter serves as the default value of the chip size Y.

✂ When you omit (δx , δy), the chip center for writing will be at array points. When you omit the parameters m and n, special writing modes will not be performed.

✂ Prior to the chip sizes you specified using CS = (csx, csy), the values specified in the pattern data are used.

4.3.7 SPPRM

(Optional; inapplicable in JBX-6000FS.)

Format:	
SPPRM	[m1], [m2], [w], [p] [, [sc] [, [ss] [, [sfs]]
	① ② ③ ④ ⑤ ⑥ ⑦
Example:	
P(1)	'jeol.j5x',MULTI,3 (1)
SPPRM	2.0,4.0,0.24,0.12,,2 (2)

- Description of the example

- (1) The preceding P command specifies the file jeol.j5x and specifies to write the chip pattern contained in that file using the triple field shift overlap writing.
- (2) For the triple field shift overlap writing, the SPPRM command sets the following conditions: maximum beam size = 2.0 μm , maximum beam area = 4.0 μm^2 , trapezoid writing slit width = 0.24 μm , trapezoid writing pitch = 0.12 μm , writing magnification = 1.0, shot-shift amount for the field shift overlap writing = half of the beam size, subfield-shift amount for the subfield-shift writing = half of the subfield size.

- Description of the command

The SPPRM command specifies shot partition parameters for each writing chip pattern data. The SPPRM command specified in a jobdeck layer block will override the SPPRM command specified in the common block if the SPPRM command in the jobdeck common block concerns the same pseudo pattern number.

For the pattern data corresponding to a pseudo pattern number, if the SPPRM command is specified neither in the jobdeck common block nor in a jobdeck layer block, the default parameter values that are provided by the system will be used.

✂ In the case of the variable shaped e-beam lithography system, shot partition parameters will be performed as beam shaping parameters. In the case of the spot e-beam lithography system, they will be performed as parameters related to the beam scanning range that can be written by a single positioning deflection.

① m1

For the variable shaped e-beam machine, this parameter is used to specify the maximum value of the longer side of the shaped rectangle beam for writing a chip.

For the spot e-beam machine, this parameter is used to specify the maximum value

of the side of the square beam scanning area that can be written by a single positioning deflection.

② m2

For the variable shaped e-beam machine, this parameter is used to specify the maximum value of the shaped rectangle beam area for writing a chip.

In the case of the spot e-beam machine, this parameter is automatically set to the square of m1.

 In the case of the spot e-beam lithography system, the specification of the parameter m2 will not be used, even if it is specified.

③ w

For the variable shaped e-beam machine, this parameter is used to specify the length of the shorter side of the shaped rectangle beam for writing a trapezoid pattern.

 For the spot e-beam machine, specification of this parameter is inapplicable.

④ p

For the variable shaped e-beam machine, this parameter is used to specify the shift pitch of the shaped rectangle beam in the shorter side direction for writing a trapezoid pattern.

 For the spot e-beam machine, specification of this parameter is not applicable.

⑤ sc

Specifies the magnification or reduction ratio of scaling for pattern writing.

If you omit sc, the magnification or reduction ratios of scaling in the directions of both X and Y axes are set to 1.0.

For the JBX-3040MV and later models, the product of the writing enlargement/reduction ratio specified in the JEOL52 (V3.1) format data and the value of this parameter is the scaling enlargement/reduction ratio to use for writing.

⑥ ss

Specifies the shift amount for partition of the shaped rectangle beam or that for the spot beam scanning area when shot shift writing is performed. The value will be the inverse of the value obtained from the maximum beam area (m2) that corresponds to the pattern shape while each size of the maximum beam area is set to be 1 both in the horizontal direction and in the vertical direction. If you omit this parameter, shot shift writing will not be performed.

If neither field fold overlap writing nor field shift overlap writing is applied to the pattern data, shot shift writing will not also be applied to it.

 For information about shot shift writing, refer to the “GENERAL STATEMENT” or “LITHOGRAPHY OPERATIONS” for the lithography system.

⑦ sfs

The parameter sfs specifies the subfield-shift amount for the subfield-shift writing. For the parameter value, specify the reciprocal of a fraction whose numerator is 1, assuming that the subfield size is 1. The subfield-shift writing is disabled by default. For the pattern data for which the overlap writing and the field shift overlap writing are disabled, the subfield-shift writing is also disabled.

 The subfield-shift writing is an optional feature for the JBX-3030MV and is a standard feature for the JBX-3040MV and later models. For details, see the manual, “General Statement” or the manual, “Operation” for your lithography system.

 Take adequate care when you specify the parameters ① to ⑦, because they are closely related to the writing precision.

4.3.8 STGPRM

(Optional/available only in the JBX-3050MV and later models)

Format:

STGPRM [a1] [, [a2] [, [a3] [, [a4], a5]]]

① ② ③ ④ ⑤

Example:

P(1) 'jeol.j5x',MULTI/FP,2
STGPRM XRASTL,XRASTL

● Description of the example

- (1) The preceding P command specifies a double field shift overlap writing in which the full pass writing applies to the chip pattern contained in the jeol.j5x file.
- (2) In that writing, the substrate stage moves in the X raster sequence from the top left of the pattern to write the pattern of the first overlap, and the substrate stage moves in the X raster sequence from the top left of the pattern to write the pattern of the second overlap.

● Description of the command

The STGPRM command specifies the method of moving the substrate stage in the full pass writing. If the preceding P command specifies the full pass writing, the STGPRM command specifies how to move the substrate stage to write the pattern in the normal writing and the nth writing in the field shift overlap writing. If the full pass writing is not specified, only the a1 parameter takes effect; the writing is performed, with no distinction between the first and second overlaps, using the specified movement method (with XZIGTL specified, the writing is performed in the usual manner).

If the P command specifies the full pass writing and if the STGPRM command is omitted, the default values specified as system parameters are used. If the P command does not specify the full pass writing and if the STGPRM command is omitted, the writing is performed in the usual manner.

① a1

Here you can specify the direction in which to move the stage for the first overlap. You can use the following parameters to determine the movement method.

Parameter	Description
XRASTL	X raster writing from the top left of the pattern
XRASTR	X raster writing from the top right of the pattern
XRASBL	X raster writing from the bottom left of the pattern
XRASBR	X raster writing from the bottom right of the pattern
XZIGTL	X zigzag writing from the top left of the pattern
XZIGTR	X zigzag writing from the top right of the pattern
XZIGBL	X zigzag writing from the bottom left of the pattern
XZIGBR	X zigzag writing from the bottom right of the pattern

You can omit this parameter; in that case, the default value specified as a system parameter is used.

② a2

Here you can specify the direction in which to move the stage for the second overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is neglected.

③ a3

Here you can specify the direction in which to move the stage for the third overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is neglected.

④ a4

Here you can specify the direction in which to move the stage for the fourth overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is neglected.

⑤ a5

Here you can specify the direction in which to move the stage for the fifth overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is neglected.

● Supplementary notes

- Omitting parameters

With both /FP and STGPRM omitted

⇒The conventional writing is performed.

With /FP omitted and STGPRM specified

⇒Only the first parameter of STGPRM takes effect, and using the movement method of that parameter, the writing is performed with no distinction between the first and second overlaps.

With /FP specified and STGPRM omitted

⇒The writing is performed using the default STGPRM value specified as a system parameter.

- Handling the SMASH writing

For a path for which SMASH is specified, the parameter that determines the movement method must be the same for all the patterns in the path. Any specification that is different will cause an error in the data-preparation stage before writing.

4.3.9 C

(Optional)

Format:

C(i) 's'

① ②

Example:

C(1) 'JEOL'

- **Description of the example**

The character string “JEOL” will be written on the substrate at the position and with attributes designated by character string number 1 of the CHAR command (CHAR/1).

- **Description of the command**

The C command specifies not only the position and attributes of a character string to be written on the substrate by designating a character string number of the C command, but also the character string to be written by giving it. The C command specified in a jobdeck layer block will override the C command specified in the common block if the C command in the common block concerns the same character string number.

① i

Character string number that specifies the position and attributes for the character string.

② s

Character string to be written on the substrate.

4.3.10 RESIST

(Optional; 【, s2】 for the JBX-6000FS, and 【, s2】 【, a】 for the JBX-6300 series, must not be omitted.)

Format:		
RESIST	s1 【,s2】 【,a】	
	① ② ③	
Example:		
RESIST	200.0,100.0	(1)
RESIST	2.4	(2)
RESIST	200.0,500.0,A	(3)
RESIST	200.0,2.4,L	(4)

- **Descriptions of the examples**

- (1) The writing uses a resist with an area sensitivity of 200.0 $\mu\text{C}/\text{cm}^2$ and a line sensitivity of 100.0 nC/cm.
- (2) The writing uses a resist with an area sensitivity of 2.4 $\mu\text{C}/\text{cm}^2$.
- (3) For writing, the JBX-6300 series and the JBX-5500 series uses a resist with an area sensitivity of 200.0 $\mu\text{C}/\text{cm}^2$ and a line sensitivity of 500.0 $\mu\text{C}/\text{cm}^2$.
- (4) For writing, the JBX-6300 series and the JBX-5500 series uses a resist with an area sensitivity of 200.0 $\mu\text{C}/\text{cm}^2$ and a line sensitivity of 2.4 nC/cm.

- **Description of the command**

The RESIST command specifies the sensitivities of the resist used for e-beam writing.

① s1

Area sensitivity of the resist; used for writing patterns with areas (rectangle and trapezoid).

- ② s2 (Effective at JBX-6000FS, JBX-6300 series and JBX-5500 series)
Line sensitivity of the resist; used for writing patterns without areas (single line, etc.).
- ③ a (Only JBX-6300 series and JBX-5500 series are effective.)
Used for specifying the unit of the line sensitivity of the resist. If this is omitted, A ($\mu\text{C}/\text{cm}^2$) is specified.

Unit of writing pattern without areas	$\mu\text{C}/\text{cm}^2$	nC/cm
Description	A	L

-  Single-line writing can be performed using the JBX-6000FS, the JBX-6300 series and the JBX-5500 series.
-  In the JBX-6000FS, the line sensitivity of the resist is represented by nC/cm. In the JBX-6000FS, the JBX-6300 series and the JBX-5500 series, s2 must not be omitted. In the JBX-9000MV and the JBX-9300FS, even if s2 is specified, it will not be applied.

4.3.11 SHOT

(Optional)

Format:	
SHOT	{ A [, s] S, t1 [, [t2] [, s]] V, r, x, y }
	① ② ③④ ⑤ ⑥⑦⑧⑨
Example:	
SHOT	A, 2 (1)
SHOT	S, 200, 500, 2 (2)
SHOT	V, 100, 100, 100 (3)

● Descriptions of the examples

- (1) The shot time (the writing time for each e-beam shot) is determined by the resist sensitivity and the beam current, and the beam scanning interval (shot pitch) of 2 is specified.
- (2) In the JBX-6000FS or JBX-6300 series or JBX-9300FS or JBX-5500 series e-beam lithography system, the shot times (200 ns for area patterns and 500 ns for single lines) and shot pitch 2 are specified.
- (3) In the JBX-9000MV or JBX-3030MV e-beam lithography system, the shot time 100ns is specified for rectangles, X trapezoids, and Y trapezoids.

● Description of the command

The SHOT command specifies the writing time for each e-beam shot (called the shot time) and also specifies the shot pitch used in the spot e-beam lithography system. The following two modes are available for determining the shot time: the auto mode that determines the shot time automatically from the resist sensitivity and the beam current, and the manual mode in which you designate the shot time according to pattern shapes.

① A

The shot time is automatically determined from the resist sensitivity and beam current.

② s

Shot pitch in a spot e-beam lithography system. If this parameter is not specified, it is set to 1. This specification is inapplicable in a variable shaped e-beam lithography system.

 When the parameter s is discarded in JBX-6000FS, the beam scanning interval specified each time writing a chip pattern in the PREAD program is effective for each writing chip.

③ S

Indicates that shot times are used in a spot e-beam lithography system.

④ t1

Shot time for area patterns in a spot e-beam lithography system.

⑤ t2

Shot time for single line patterns in a spot e-beam lithography system.

 The parameter t2 must be specified in JBX-6000FS, JBX-6300 series. and JBX-5500 series However, in JBX-9300FS, it is discarded, even if it is specified.

⑥ V

Indicates that the shot times are used in a variable shaped e-beam lithography system.

⑦ r

Shot time for rectangle patterns in a variable shaped e-beam lithography system.

⑧ x

Shot time for X trapezoid patterns in a variable shaped e-beam lithography system.

⑨ y

Shot time for Y trapezoid patterns in a variable shaped e-beam lithography system.

4.3.12 MODULAT

(Optional)

Format:		
t:	MODULAT	{((r,v) [,...])
①		②③
–		((r0,r1,r2),(v0,v1,v2)) [,...])}
		④⑤⑥ ⑦ ⑧ ⑨
or		
't':	MODULAT	{((r,v) [,...])
–		((r0,r1,r2),(v0,v1,v2)) [,...])}
Examples:		
MDLT01:	MODULAT	((0,–10)) (1)
MDLT02:	MODULAT	((1,5,9),(–30,10,45))
–		,((10,15,20),(–95,–50,3100))) (2)

● Descriptions of the examples

- (1) Shot time modulation of –10% is added to the shot time for patterns with a shot rank number 0.
- (2) Shot time modulation is allotted to the shot rank numbers 1 to 9, increasing geometrically by 10%, with the shot time modulation at the shot rank 5 being the reference point (0%), and to the shot rank numbers 10 to 20, also increasing geometrically (but by –50%), with the shot time modulation at the shot rank 15 being the reference point. Shot time modulation thus allotted is added to patterns having a corresponding shot rank number.

● Description of the command

The MODULAT command specifies modulation of shot time (defined by the SHOT command) for each shot rank of chip pattern data. Shot time modulation may be allotted individually to each shot rank number, or to shot rank numbers in a specific range at a certain fixed ratio.

Correspondence between chip pattern writing with shot time modulation and the MODULAT command is made by allotting a shot time modulation table name defined by the ASSIGN command in the common block of the jobdeck.

The shot time modulation of 0% will be allotted for shot rank numbers that are not specified by this command.

The shot time during e-beam writing is obtained from the specified shot time (St) and shot time modulation (V) in the following equation:

$$(\text{Shot time during writing}) = \text{St} \times \frac{100 + V}{100}$$

- ① t
Name of the shot time modulation table. If the character string that indicates the shot time modulation table name begins with a character other than a letter, enclose the entire character string in single quotes ('').*
- ② r
Shot rank number.
- ③ v
Shot time modulation amount.
- ④ r0
First shot rank number to modify.
- ⑤ r1
Shot rank number where shot time modulation is 0.
- ⑥ r2
Last shot rank number to modify.
- ⑦ v0
The lower limit of shot time modulation.
- ⑧ v1
Change (ratio) of shot time modulation.
- ⑨ v2
Upper limit of shot time modulation.

 By specifying r0 to r2 and v0 to v2, the shot time modulation (V) for an arbitrary shot rank number (R) is determined from the following equation:

$$V = \left\{ \left(1 + \frac{v1}{100} \right)^{(R-r1)} - 1 \right\} \times 100$$

The parameters r0 and r2, and v0 and v2 are limited as follows:

$$r0 \leq r2, \text{ and } v0 \leq v2$$

* This description is available only for the JBX-3030MV.

4.3.13 EOS

(Optional; applicable only in JBX-6000FS, JBX-6000FS/E, JBX-6300 series and JBX-5500 series.)

Format:		
EOS	m,{t 'c'}	
	① ② ③	
Example:		
EOS	2,4	(1)
EOS	2,'DEFAULT'	(2)

● Description of the example

- (1) E-beam writing is carried out using the e-beam writing and calibration (CALIB) conditions given by EOS mode number 2 and EOS table number 4.
- (2) Writing is carried out using the e-beam writing and calibration (CALIB) conditions given by EOS mode number 2 and calibration condition name “DEFAULT”.

● Description of the command

The EOS command specifies hardware correction parameters for e-beam writing and calibration conditions by using a combination of the EOS mode, and the EOS table or the calibration condition name.

Using the calibration (CALIB) program, various correction factors for the hardware of the e-beam lithography system are sorted according to their writing conditions and calibration conditions, and stored in the storage for the EOS modes and the EOS tables or the calibration conditions, respectively. If EOS statements are given in both the cassette block of the schedule file and the layer block of the jobdeck file, the specification in the schedule file will override that in the jobdeck file.

① m

EOS mode number.

② t

EOS table number.

 Specify the EOS table number in JBX-6000FS.

③ c

Calibration condition name.

 specify the calibration condition name in JBX-6000FS/E, JBX-6300 series and JBX-5500 series.

 The EOS command must be given either in jobdeck layer blocks or in schedule cassette blocks.

4.3.14 DEFOCUS

(Optional)

Format:	
DEFOCUS	d ①
Example:	
DEFOCUS	-1000

- **Description of the example**

The focus value of the electron optical system of the e-beam lithography system is reduced from the present setting by 1000 points.

- **Description of the command**

The DEFOCUS command adjusts the focus value of the electron optical system by the specified defocus value. To perform e-beam writing by applying the GHOST™* technique, a procedure to correct for the proximity effect, the focus is deliberately changed. If this command is not given, the defocus value 0 will be specified.

① d

Amount of adjustment to focus setting.

4.3.15 SCALEB

(Optional; applicable only in JBX-6000FS.)

Format:	
SCALEB	bx[,by] ① ②
Example:	
SCALEB	0.05

- **Description of the example**

The offset amounts (bias values) of the scaling ratio in fine pitch control writing are 0.05 in both the X and Y directions.

☞ For information about fine pitch control writing, refer to the “General Statement” or “Lithography Operations” for the lithography system.

- **Description of the command**

The SCALEB command indicates the offset amounts of the scaling ratios in fine pitch control writing. The specified offset amounts are to be added to the scaling ratio of the fine pitch control writing specified by the SCALE command. This command is used as a remark for setting the hardware of the e-beam lithography system.

* GHOST™ is a registered trademark of Hewlett-Packard Company.

- ① bx
Indicates the offset amount of scaling ratio (in X direction) used in fine pitch control writing.
- ② by
Indicates the offset amount of scaling ratio (in Y direction) used in fine pitch control writing. If you omit this parameter, the system will use the value of bx.

4.3.16 OBJAPT

(Optional; inapplicable in JBX-9000MV and JBX-3030MV)

Format:	
OBJAPT	a ①
Example:	
OBJAPT	2

- **Description of the example**

The objective lens aperture number 2 is to be used for e-beam writing.

- **Description of the command**

The OBJAPT command indicates the objective lens (OL) aperture to be used for e-beam writing. This command is used as a remark for setting the hardware of the e-beam lithography system.

- ① a
Indicates the aperture number of the OL aperture to be used for e-beam writing.

4.3.17 STDCUR

(Optional; inapplicable in JBX-9000MV and JBX-3030MV)

Format:	
STDCUR	c ①
Example:	
STDCUR	2

- **Description of the example**

The beam current of 2 nA is to be used for e-beam writing.

- **Description of the command**

The STDCUR command indicates the level of beam current to be used for e-beam writing. This command is used as a remark for setting the hardware of the e-beam lithography system.

① c

Indicates the level of the beam current to be used for e-beam writing.

4.3.18 RESTYP

(Optional)

Format:	
RESTYP	t[, 'n']
	① ②
Example:	
RESTYP	POSI, 'PMMA'

- **Description of the example**

The resist on the substrate is positive PMMA.

- **Description of the command**

The RESTYP command indicates the type and name of the resist coated on the substrate. This command is used as a remark for setting the hardware of the e-beam lithography system.

① t

Indicates the type of the resist coated on the substrate.

② n

Indicates the name of the resist coated on the substrate.

4.3.19 PWIDTH

(Optional)

Format:	
PWIDTH	w
	①
Example:	
PWIDTH	0.10

- **Description of the example**

The chip pattern size (line width) is 0.10 μm .

- **Description of the command**

The PWIDTH command indicates the minimum line width or the critical dimension (CD) of the pattern width for the chip pattern that will be the basis for setting the writing condition. This command is used as a remark for setting the hardware of the e-beam lithography system.

① w

Indicates the size (line width) of the chip pattern to be used as the base for setting writing conditions.

4.4 Termination Command

4.4.1 END

(Required)

Format: END (no parameters)

- **Description of the command**

The END command ends the jobdeck. Any lines (records) after this command will not affect the jobdeck.

4.5 Control Characters

4.5.1 Hyphen (-)

(Optional)

Format:	
-	Parameter list (continued)
Example:	
ASSIGN	P(1) ->((1,1),(2,2) (1)
-	,(3,3),(4,4),...) (2)

- **Description of the example**

When a command and the parameter list that follows are too long to write in one physical record, the parameter list can be divided into the initial line (1) and the continuation line (2) as shown above. In the above, the ASSIGN command is used as an example, but the parameter lists of other commands can be continued using the hyphen as well.

- **Description of the control character**

The hyphen (-) placed in the first column of a record indicates that a parameter list is continued from the immediately previous line. That hyphen (-) must always be placed in the first column of the next record. This control character can be used in both common and layer blocks of the jobdeck file.

4.5.2 Equal sign (=)

(Optional)

Format:		
=	Parameter list	
Example:		
ASSIGN	P(1) ->((1,1),(2,2))	(1)
=	P(2) ->((3,3),(4,4))	(2)

- **Description of the example**

Instead of repeating the same command, the second command is expressed by substituting an equal sign (=) as shown in (1) and (2) above. In the above, the ASSIGN command is used as an example, but other commands can be repeated using the equal sign.

- **Description of the control character**

An equal sign (=) placed in the first column in a line behaves as a command identical to the one that is given in the immediately previous line. That equal sign (=) must always be placed in the first column of the next record. This control character can be used in both common and layer blocks of the jobdeck file.

4.5.3 Semicolon (;)

(Optional)

Format:		
;	Character string representing a comment	
Example:		
ASSIGN	P(1) ->((1,1)) ; FOR DUMMY	

- **Description of the example**

The comment, FOR DUMMY, is added to the ASSIGN command after the semicolon (;).

- **Description of the control character**

The text from the semicolon (;) to the end of the line (record) is treated as a comment. The comment does not affect the jobdeck file. It can be used in both common and layer blocks of the jobdeck file.

4.5.4 At mark (@)

(Optional)

Format:

@ 'f'

①

Example:

@ 'include.inc'

- **Description of the example**

The file named “include.inc” is inserted in the record marked by the at symbol (@).

- **Description of the symbol**

An at mark (@) in a record designates that the specific file named by this statement is to be inserted in the record. The at mark must always be placed in the first column of the record and the jobdeck must not have any contradiction after the insertion of the file. This statement can be used in both common and layer blocks of the jobdeck file.

① f

Specifies the name of the file to be inserted.

5 APPENDICES

5.1 Limits of Commands

5.1.1 Maximum number of commands that can be repeated

**Table 6 Maximum number of commands that can be repeated
(Models except JBX-3030MV and JBX-5500 series)**

Definition	Command	Limit	Unit
Writing path	PATH : PEND	32	/jobdeck
Chip array	ARRAY : AEND	99	/writing path definition
	ASSIGN	No limit	/chip array definition
	SKIP	No limit	/chip array definition
Character string position	CHAR	8*	/jobdeck
Writing pattern	P	99	/layer
Writing character string	C	4	/layer
Writing condition	MODULAT	99	/layer

**Table 7 Maximum number of commands that can be repeated
(JBX-3030MV)**

Definition	Command	Limit	Unit
Writing path	PATH : PEND	2000**	/jobdeck
Chip array	ARRAY : AEND	2000	/jobdeck
	ASSIGN	No limit	/chip array definition
	SKIP	No limit	/chip array definition
Character string position	CHAR	8*	/jobdeck
Writing pattern	P	2000***	/layer
Writing character string	C	4	/layer
Writing condition	MODULAT	2000	/layer

* Character string number i cannot be duplicated in a jobdeck.

** For the JBX-3030MV, the maximum might be limited to 256 depending on the hardware configuration. For the JBX-3040MV and later models, the maximum is 2000.

*** For the JBX-3030MV, the maximum might be limited to 150 or 300 depending on the hardware configuration. For the JBX-3040MV and later models, the maximum is 2000.

**Table 8 Maximum number of commands that can be repeated
(JBX-5500 series)**

Definition	Command	Limit	Unit
Writing path	PATH : PEND	32	/jobdeck
Chip array	ARRAY : AEND	2000	/jobdeck
	ASSIGN	No limit	/chip array definition
	SKIP	No limit	/chip array definition
Character string position	CHAR	8*	/jobdeck
Writing pattern	P	99	/layer
Writing character string	C	4	/layer
Writing condition	MODULAT	1000	/layer

5.1.2 Maximum number of nesting levels in chip array definitions (assigned by the ASSIGN command)

They are a maximum of 10 levels.

5.1.3 Multiple specifications by the same command

The latter specification always overrides the previous one in the common block, also in each layer block.

* Character string number i cannot be duplicated in a jobdeck.

5.2 Limits of Parameters

5.2.1 JBX-6000FS

Terminology	
min:	minimum value
max:	maximum value
step:	step
unit:	unit
slct:	selectable value
fix:	fixed value

Table 9 Limits of parameters for JBX-6000FS

Command	Parameter	Limits	Remarks (JBX-6000FS)
JOB	/W	fix = /W	
	j	Up to 9 uppercase letters, numbers, and symbols. Must start with a letter.	Character string containing single quote (') cannot be specified.
	d1	min = 2.0 max = 6.0 step = 0.1 unit= inch	max = 4.0 in JBX-5DII(U)
	d2	min = 6.0 (inches) or -152.400 (mm) max= 6.0 (inches) or 152.400 (mm) step = 0.1 (inches) or 0.001 (mm) unit= inch or mm	
	a	slct = I or M	
	b	slct = I or O	
OPTION	t	fix = T	15-character string that designates date and time: dd-mmm-yy, hh:mm.
	i	fix = M	16-character string that designates model name.
	p	slct = X or Y	
	c	slct = X or Y	
	sf	Inapplicable	
DEVINF	d	Up to 16 writable uppercase letters, numbers, and symbols	
GLMPOS	px	min = -76200.000 max = 76200.000 step = 0.001 unit = μ m	
	py	Same as above	

Command	Parameter	Limits	Remarks (JBX-6000FS)
	qx	min = px max = 76200.000 step = 0.001 unit = μm	
	qy	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
	rx	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
	ry	min = py max = 76200.000 step = 0.001 unit = μm	
	sx	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
	sy	min = -76200.000 max = py step = 0.001 unit = μm	
GLMP	w	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	l	min = w/2 max = 152400.000 step = 0.001 unit = μm	
	t	slct = 0 or 1	
	f	slct = 0, 1, 2, or 3	
GLMQRS	w	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	l	min = w/2 max = 152400.000 step = 0.001 unit = μm	
	t	slct = 0 or 1	
	f	slct = 0, 1, 2, or 3	
PLMPOS	xn	Inapplicable	
	yn	Inapplicable	

Command	Parameter	Limits	Remarks (JBX-6000FS)
PLMARK	w	Inapplicable	
	l	Inapplicable	
PATH	/P	Inapplicable	
	/S	Inapplicable	
	p	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
SCALE	sx	min = 0.970000 max = 1.030000 step = 0.000001	
	sy	Same as above	
ARRAY	a	min = 1 max = 2000 step = 1	
	x	min = -104900.000 max = 76200.000 step = 0.001 unit = μm	*
	m	min = 1 max = 255 step = 1	
	p	min = 0.000 max = 152400.000 step = 0.001 unit = μm	
	y	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
	n	min = 1 max = 255 step = 1	
	q	min = 0.000 max = 152400.000 step = 0.001 unit = μm	
CHMPOS	x1	min = -152400.000 max = 152400.000 step = 0.001 unit = μm	
	y1	Same as above	
	x2	Same as above	x2≠x1

* When the position of a chip array is mirrored with respect to the Y axis according to the specification(s) of the OPTION command in the common block and/or the MIRROR command in a layer block, the limits of the parameter x are changed as follows:
min = -76200.000, max = 104900.000, step = 0.001, unit = μm .

Command	Parameter	Limits	Remarks (JBX-6000FS)
	y2	min = $-(x_2 - x_1) \times \tan 45^\circ + y_1$ max = $(x_2 - x_1) \times \tan 45^\circ + y_1$ step = 0.001 unit = μm	
	x3	min = $-(y_3 - y_1) \times \tan 45^\circ + x_1$ max = $(y_3 - y_1) \times \tan 45^\circ + x_1$ step = 0.001 unit = μm	
	y3	min = -152400.000 max = 152400.000 step = 0.001 unit = μm	y3 ≠ y1
	x4	Inapplicable	
	y4	Inapplicable	
CHMARK	w	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	l	min = w/2 max = 152400.000 step = 0.001 unit = μm	
	t	slct = 0 or 1	
	f	slct = 0, 1, 2, or 3	
ASSIGN	i	min = 1 max = 99 step = 1	
	a	Same as above	
	j	min = 1 max = 255 step = 1	Asterisk (*) can be used.
	k	Same as above	Same as above
	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	t1	Inapplicable	
SKIP	j	min = 1 max = 255 step = 1	Asterisk (*) can be used.
	k	Same as above	Same as above
CHAR	i	slct = 1, 2, 3, 4, T, M, D, or L	
	x	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
	y	Same as above	

Command	Parameter	Limits	Remarks (JBX-6000FS)
	a	slct = 0 or 1	
	b	slct = 0 or 1	
	m	Inapplicable	
	r	Inapplicable	
P	i	min = 1 max = 99 step = 1	
	p	Up to 24 uppercase letters, numbers, and symbols that can be used as a pattern ID name.	ID name containing single quote (') cannot be specified
	δx	min = -152400.000 max = 152400.000 step = 0.001 unit = μm	
	δy	Same as above	
	m	Inapplicable	
	n	Inapplicable	
	csx	Inapplicable	
	csy	Inapplicable	
SPPRM	m1	Inapplicable	
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	Inapplicable	
	ss	Inapplicable	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	*

* The character size is 1.6 mm $\times$ 1.6 mm, and the writing pitch is 1.6 mm.

Command	Parameter	Limits	Remarks (JBX-6000FS)
LAYER	/P	Inapplicable	
	n	min = 1 max = 99 step = 1	
	s	Up to 16 writable uppercase letters, numbers, and symbols.	
CIRCLE	d2	min = - 6.0 (inches) or -152.400 (mm) max = 6.0 (inches) or 152.400 (mm) step = 0.1 (inches) or 0.001 (mm) unit = inch or mm	
	a	slct = I or M	
	b	slct = I or O	
CANCEL	p	slct = T, M or D	
MIRROR	p	slct = X or Y	
	c	slct = X or Y	
SCALE	sx	min = 0.970000 max = 1.030000 step = 0.000001	
	sy	Same as above	
P	i	min = 1 max = 99 step = 1	
	p	Up to 24 uppercase letters, numbers, and symbols that can be used as a pattern ID name.	Character string containing single quote (') cannot be specified.
	δx	min = -152400.000 max = 152400.000 step = 0.001 unit = μm	
	δy	Same as above	
	m	Inapplicable	
	n	Inapplicable	
	csx	Inapplicable	
	csy	Inapplicable	
SPPRM	m1	Inapplicable	
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	Inapplicable	
	ss	Inapplicable	
	sfs	Inapplicable	

Command	Parameter	Limits	Remarks (JBX-6000FS)
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	*
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	min = 0.001 max = 1000.000 step = 0.001 unit = nC/cm	
	a	Inapplicable	
SHOT	s	min = 1 max = 8 step = 1	**
	t1	min = 167 max = 3900000 step = 1 unit = ns	
	t2	Same as above	
	r	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
MODULAT	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	r	min = 0 max = 63 step = 1	In single line writing, 16 and up are ignored.
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	

* The character size is 1.6 mm × 1.6 mm, and the writing pitch is 1.6 mm.

** In JBX-5DII (2 MHz), max = 40; in both JBX-5DII (6 MHz) and JBX-5000LS (6 MHz), max = 20; in both JBX-5FE (6 MHz) and JBX-6000FS (12 MHz), max = 8.

Command	Parameter	Limits	Remarks (JBX-6000FS)
	r0	min = 0 max = r2 step = 1	In single line writing, 16 and up are ignored.
	r1	min = 0 max = 63 step = 1	In single line writing, 16 and up are ignored.
	r2	min = r0 max = 63 step = 1	In single line writing, 16 and up are ignored.
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
EOS	m	min = 1 max = 8 step = 1	
	t	min = 0 max = 9 step = 1	
	c	Inapplicable	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present. It does not work even if you specify this.
SCALEB	bx	min = -0.070000 max = 0.070000 step = 0.000001	
	by	Same as above	
OBJAPT	a	min = 1 max = 3 step = 1	max = 5 in JBX-5DII
STDCUR	c	min = 0.001 max = 100.000 step = 0.001 unit = nA	
RESTYP	t	slct = POSI or NEGA	
	n	Up to 20 alphanumeric characters and symbols.	Character string containing single quote (') cannot be specified.

Command	Parameter	Limits	Remarks (JBX-6000FS)
PWIDTH	w	min = 0.010 max = 100.000 step = 0.001 unit = μm	
@	f	Character string consisting of any alphanumeric characters and symbols allowed in the computer system as a filename.	Character string containing single quote (') cannot be specified.

5.2.2 JBX-9000MV

Terminology	
min:	minimum value
max:	maximum value
step:	step
unit:	unit
slct:	selectable value
fix:	fixed value

Table 10 Limits of parameters for JBX-9000MV

Command	Parameter	Limits	Remarks (JBX-9000MV)
JOB	/W	fix = /W	
	j	Up to 9 uppercase letters, numbers, and symbols. Must start with a letter.	Character string containing single quote (') cannot be specified.
	d1	min = 2.0 max = 9.0 step = 0.1 unit= inch	
	d2	min = -9.0 (inches) or -230.000 (mm) max= 9.0 (inches) or 230.000 (mm) step = 0.1 (inches) or 0.001 (mm) unit= inch or mm	
	a	slct = I or M	
	b	slct = I or O	
OPTION	t	fix = T	20-character string that designates date and time: dd-mm-yy, hh:mm:ss.
	i	fix = M	16-character string that designates model name.
	p	slct = X or Y	
	c	slct = X or Y	
	sf	Inapplicable	
DEVINF	d	Up to 16 writable uppercase letters, numbers, and symbols	
GLMPOS	px	min = -115000.000 max = 115000.000 step = 0.001 unit = μ m	
	py	Same as above	
	qx	min = px max = 115000.000 step = 0.001 unit = μ m	

Command	Parameter	Limits	Remarks (JBX-9000MV)
	qy	min = -115000.000 max = 115000.000 step = 0.001 unit = μm	
	rx	Inapplicable	
	ry	Inapplicable	
	sx	Inapplicable	
	sy	Inapplicable	
GLMP	w	min = 0.000 max = 50.000 step = 0.001 unit = μm	
	l	min = w/2 max = 115000.000 step = 0.001 unit = μm	
	t	Inapplicable	
	f	Inapplicable	
GLMQRS	w	min = 0.000 max = 50.000 step = 0.001 unit = μm	
	l	min = w/2 max = 115000.000 step = 0.001 unit = μm	
	t	Inapplicable	
	f	Inapplicable	
PLMPOS	xn	min = -115000.000 max = 115000.000 step = 0.001 unit = μm	
	yn	Same as above	
PLMARK	w	min = 0.000 max = 50.000 step = 0.001 unit = μm	
	l	min = w/2 max = 115000.000 step = 0.001 unit = μm	
PATH	/P	fix = /P	
	/S	fix = /S	
	p	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.

Command	Parameter	Limits	Remarks (JBX-9000MV)
SCALE	sx	Inapplicable	
	sy	Inapplicable	
ARRAY	a	min = 1 max = 99 step = 1	
	x	min = -140000.000 max = 120000.000 step = 0.001 unit = μm	*
	m	min = 1 max = 255 step = 1	
	p	min = 0.000 max = 260000.000 step = 0.001 unit = μm	
	y	min = -120000.000 max = 120000.000 step = 0.001 unit = μm	
	n	min = 1 max = 255 step = 1	
	q	min = 0.000 max = 240000.000 step = 0.001 unit = μm	
CHMPOS	x1	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	
	y1	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	
	x2	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	$x2 - x1 > 0$
	y2	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	

* When the position of a chip array is mirrored with respect to the Y axis according to the specification(s) of the OPTION command in the common block and/or the MIRROR command in a layer block, the limits of the parameter x are changed as follows:
min = -120000.000, max = 140000.000, step = 0.001, unit = μm .

Command	Parameter	Limits	Remarks (JBX-9000MV)
	x3	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	$x3 - x1 > 0$
	y3	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	$y1 - y3 > 0$ & $y2 - y3 > 0$
	x4	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	$x3 - x4 > 0$ & $x2 - x4 > 0$
	y4	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	$y1 - y4 > 0$ & $y2 - y4 > 0$
CHMARK	w	min = 0.000 max = 50.000 step = 0.001 unit = μm	
	l	min = w/2 max = 115000.000 step = 0.001 unit = μm	
	t	Inapplicable	
	f	Inapplicable	
ASSIGN	i	min = 1 max = 99 step = 1	
	a	Same as above	
	j	min = 1 max = 255 step = 1	Asterisk (*) can be used.
	k	Same as above	Same as above
	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	t1	Inapplicable	
SKIP	J	min = 1 max = 255 step = 1	Asterisk (*) can be used.
	k	Same as above	Same as above
CHAR	i	slct = 1, 2, 3, 4, T, M, D, or L	
	x	min = -140000.000 max = 120000.000 step = 0.001 unit = μm	

Command	Parameter	Limits	Remarks (JBX-9000MV)
	y	min = -120000.000 max = 120000.000 step = 0.001 unit = μm	
	a	slct = 0 or 1	
	b	slct = 0 or 1	
	m	Inapplicable	
	r	Inapplicable	
P	i	min = 1 max = 99 step = 1	
	p	Character string consisting of up to 24 characters (any alphanumeric characters and symbols allowed in the computer system as a filename).	Character string containing single quote (') cannot be specified.
	δx	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	
	δy	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	
	m	slct = MULTI, GATHER, or OVERLAP	
	n	min = 2 max = 10 step = 1	Only max = 5 is applicable in m = MULTI, and only max = 8 is applicable in m = OVERLAP.
	csx	Inapplicable	
	csy	Inapplicable	
SPPRM	m1	min = 0.002 max = 4.000 step = 0.002 unit = μm	* In the case of 50 kV of accelerating voltage
	m2	min = 0.000004 max = 16.000000 step = 0.000004 unit = μm^2	** In the case of 50 kV of accelerating voltage
	w	min = 0.008 max = 2.040 step = 0.008 unit = μm	*** In the case of 50 kV of accelerating voltage The values should be multiples of p.

* In the case of 25 kV accelerating voltage, min = 0.004, max = 4.000, and step = 0.004.

** In the case of 25 kV accelerating voltage, min = 0.000016, max = 16.000000, and step = 0.000016.

*** In the case of 25 kV accelerating voltage, min = 0.016, max = 4.080, and step = 0.016.

Command	Parameter	Limits	Remarks (JBX-9000MV)
	p	min = 0.008 max = 2.040 step = 0.008 unit = μm	In the case of 50 kV of accelerating voltage*
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	**
LAYER	/P	fix = /P	
	n	min = 1 max = 100 step = 1	
	s	Up to 16 writable uppercase letters, numbers, and symbols.	
CIRCLE	d2	min = -9.0 (inches) or -230.000 (mm) max = 9.0 (inches) or 230.000 (mm) step = 0.1 (inches) or 0.001 (mm) unit = inch or mm	
	a	slct = I or M	
	b	slct = I or O	
CANCEL	p	slct = T, M, or D	
MIRROR	p	slct = X or Y	
	c	slct = X or Y	
SCALE	sx	Inapplicable	
	sy	Inapplicable	

* In the case of 25 kV accelerating voltage, min = 0.016, max = 4.080, and step = 0.016.

** The character size is 1.0 mm $\times$ 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-9000MV)
P	i	min = 1 max = 99 step = 1	
	p	Character string consisting of up to 24 characters (any alphanumeric characters and symbols allowed in the computer system as a filename).	Character string containing single quote (') cannot be specified.
	δx	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	
	δy	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	
	m	slct = MULTI, GATHER, or OVERLAP	
	n	min = 2 max = 10 step = 1	Only max = 5 is applicable in m = MULTI, and only max = 8 is applicable in m = OVERLAP.
	csx	Inapplicable	
	csy	Inapplicable	
SPPRM	m1	min = 0.002 max = 4.000 step = 0.002 unit = μm	* In the case of 50 kV of accelerating voltage
	m2	min = 0.000004 max = 16.000000 step = 0.000004 unit = μm^2	** In the case of 50 kV of accelerating voltage
	w	min = 0.008 max = 2.040 step = 0.008 unit = μm	*** In the case of 50 kV of accelerating voltage The values should be multiples of p.
	p	min = 0.008 max = 2.040 step = 0.008 unit = μm	*** In the case of 50 kV of accelerating voltage
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	

* In the case of 25 kV accelerating voltage, min = 0.004, max = 4.000, and step = 0.004.

** In the case of 25 kV accelerating voltage, min = 0.000016, max = 16.000000, and step = 0.000016.

*** In the case of 25 kV accelerating voltage, min = 0.016, max = 4.080, and step = 0.016.

Command	Parameter	Limits	Remarks (JBX-9000MV)
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	*
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	Inapplicable	
	a	Inapplicable	
SHOT	s	Inapplicable	
	t1	Inapplicable	
	t2	Inapplicable	
	r	min = 20 max = 16382000 step = 5 unit = ns	
	x	Same as above	
	y	Same as above	
MODULAT	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	r	min = 0 max = 63 step = 1	
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r0	min = 0 max = r2 step = 1	
	r1	min = 0 max = 63 step = 1	

* The character size is 1.0 mm × 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-9000MV)
	r2	min = r0 max = 63 step = 1	
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
EOS	m	Inapplicable	
	t	Inapplicable	
	c	Inapplicable	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present. It does not work even if you specify this.
SCALEB	bx	Inapplicable	
	by	Inapplicable	
OBJAPT	a	Inapplicable	
STDCUR	c	Inapplicable	
RESTYP	t	slct = POSI or NEGA	
	n	Up to 20 alphanumeric characters and symbols.	Character string containing single quote (') cannot be specified.
PWIDTH	w	min = 0.010 max = 100.000 step = 0.001 unit = μm	
@	f	Character strings consisting of any alphanumeric characters and symbols allowed in the computer system as a filename.	Character string containing single quote (') cannot be specified.

5.2.3 JBX-9300FS

Terminology	
min:	minimum value
max:	maximum value
step:	step
unit:	unit
slct:	selectable value
fix:	fixed value

Table 11 Limits of parameters for JBX-9300FS

Command	Parameter	Limits	Remarks (JBX-9300FS)
JOB	/W	fix = /W	
	j	Up to 9 uppercase letters, numbers, and symbols. Must start with a letter.	Character string containing single quote (') cannot be specified.
	d1	min = 2.0 max = 9.0 step = 0.1 unit= inch	*
	d2	min = – 12.0 (inches) or –300.000 (mm) max= 12.0 (inches) or 300.000 (mm) step = 0.1 (inches) or 0.001 (mm) unit= inch or mm	
	a	slct = I or M	
	b	slct = I or O	
OPTION	t	fix = T	15-character string that designates date and time: dd-mmm-yy, hh:mm.
	i	fix = M	16-character string that designates model name.
	p	slct = X or Y	
	c	slct = X or Y	
	sf	Inapplicable	
DEVINF	d	Up to 16 writable uppercase letters, numbers, and symbols	
GLMPOS	px	min = –115000.000 max = 115000.000 step = 0.001 unit = μm	

* The maximum value for a circular workpiece such as a wafer is 8.0. To perform pattern writing of a 12-inch substrate, an additional option is required.

Command	Parameter	Limits	Remarks (JBX-9300FS)
	py	Same as above	
	qx	min = px max = 115000.000 step = 0.001 unit = μm	
	qy	min = -115000.000 max = 115000.000 step = 0.001 unit = μm	
	rx	Inapplicable	
	ry	Inapplicable	
	sx	Inapplicable	
	sy	Inapplicable	
GLMP	w	min = 0.000 max = 50.000 step = 0.001 unit = μm	
	l	min = w/2 max = 115000.000 step = 0.001 unit = μm	
	t	slct = 0 or 1	
	f	slct = 0, 1, 2, or 3	
GLMQRS	w	min = 0.000 max = 50.000 step = 0.001 unit = μm	
	l	min = w/2 max = 115000.000 step = 0.001 unit = μm	
	t	slct = 0 or 1	
	f	slct = 0, 1, 2, or 3	
PLMPOS	xn	Inapplicable	
	yn	Inapplicable	
PLMARK	w	Inapplicable	
	l	Inapplicable	
PATH	/P	Inapplicable	
	/S	Inapplicable	
	p	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
SCALE	sx	min = 0.975000 max = 1.025000 step = 0.000001	

Command	Parameter	Limits	Remarks (JBX-9300FS)
	sy	Same as above	
ARRAY	a	min = 1 max = 99 step = 1	
	x	min = -140000.000 max = 120000.000 step = 0.001 unit = μm	*
	m	min = 1 max = 255 step = 1	
	p	min = 0.000 max = 260000.000 step = 0.001 unit = μm	
	y	min = -120000.000 max = 120000.000 step = 0.001 unit = μm	
	n	min = 1 max = 255 step = 1	
	q	min = 0.000 max = 240000.000 step = 0.001 unit = μm	
CHMPOS	x1	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	
	y1	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	
	x2	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	$x2 - x1 > 0$
	y2	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	

* When the position of a chip array is mirrored with respect to the Y axis according to the specification(s) of the OPTION command in the common block and/or the MIRROR command in a layer block, the limits of the parameter x are changed as follows:
min = -120000.000, max = 140000.000, step = 0.001, unit = μm .

Command	Parameter	Limits	Remarks (JBX-9300FS)
	x3	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	$x3 - x1 > 0$
	y3	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	$y1 - y3 > 0$ & $y2 - y3 > 0$
	x4	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	$x3 - x4 > 0$ & $x2 - x4 > 0$
	y4	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	$y1 - y4 > 0$ & $y2 - y4 > 0$
CHMARK	w	min = 0.000 max = 50.000 step = 0.001 unit = μm	
	l	min = $w/2$ max = 115000.000 step = 0.001 unit = μm	
	t	slct = 0 or 1	
	f	slct = 0, 1, 2, or 3	
ASSIGN	i	min = 1 max = 99 step = 1	
	a	Same as above	
	j	min = 1 max = 255 step = 1	Asterisk (*) can be used.
	k	Same as above	Same as above
	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	t1	Inapplicable	
SKIP	j	min = 1 max = 255 step = 1	Asterisk (*) can be used.
	k	Same as above	Same as above
CHAR	i	slct = 1, 2, 3, 4, T, M, D, or L	
	x	min = -140000.000 max = 120000.000 step = 0.001 unit = μm	

Command	Parameter	Limits	Remarks (JBX-9300FS)
	y	min = -120000.000 max = 120000.000 step = 0.001 unit = μm	
	a	slct = 0 or 1	
	b	slct = 0 or 1	
	m	Inapplicable	
	r	Inapplicable	
P	i	min = 1 max = 99 step = 1	
	p	Character string consisting of up to 24 characters (any alphanumeric characters and symbols allowed in the computer system as a filename).	Character string containing single quote (') cannot be specified.
	δx	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	
	δy	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	
	m	slct = MULTI, GATHER, or OVERLAP	
	n	min = 2 max = 10 step = 1	In the case of m = MULTI or m = OVERLAP, max = 5 or max = 8, respectively.
	csx	Inapplicable	
	csy	Inapplicable	
SPPRM	m1	min = 0.002 max = 8.000 step = 0.002 unit = μm	In the case of 50 kV of accelerating voltage*
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	

* In the case of 25 kV accelerating voltage, min = 0.004, max = 16.000, and step = 0.004.

In the case of 100 kV accelerating voltage, min = 0.001, max = 4.000, and step = 0.001.

The JBX-9300FS does not include the 25 kV accelerating voltage unit in the basic composition. To use 25 kV accelerating voltage, an additional option is required.

Command	Parameter	Limits	Remarks (JBX-9300FS)
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	*
LAYER	/P	Inapplicable	
	n	min = 1 max = 99 step = 1	
	s	Up to 16 writable uppercase letters, numbers, and symbols.	
CIRCLE	d2	min = - 12.0 (inches) or -300.000 (mm) max = 12.0 (inches) or 300.000 (mm) step = 0.1 (inches) or 0.001 (mm) unit = inch or mm	
	a	slct = I or M	
	b	slct = I or O	
CANCEL	p	slct = T, M, or D	
MIRROR	p	slct = X or Y	
	c	slct = X or Y	
SCALE	sx	min = 0.975000 max = 1.025000 step = 0.000001	
	sy	Same as above	
P	i	min = 1 max = 99 step = 1	
	p	Character string consisting of up to 24 characters (any alphanumeric characters and symbols allowed in the computer system as a filename).	Character string containing single quote (') cannot be specified.
	δx	min = -260000.000 max = 260000.000 step = 0.001 unit = μm	

* The character size is 1.0 mm × 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-9300FS)
	δy	min = -240000.000 max = 240000.000 step = 0.001 unit = μm	
	m	slct = MULTI, GATHER, or OVERLAP	
	n	min = 2 max = 10 step = 1	In the case of m = MULTI or m = OVERLAP, max = 5 or max = 8, respectively.
	csx	Inapplicable	
	csy	Inapplicable	
SPPRM	m1	min = 0.002 max = 8.000 step = 0.002 unit = μm	In the case of 50 kV of accelerating voltage*
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	**
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	Inapplicable	
	a	Inapplicable	

* In the case of 100 kV accelerating voltage, min = 0.001, max = 4.000, and step = 0.001.

** The character size is 1.0 mm $\times$ 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-9300FS)
SHOT	s	min = 1 max = 128 step = 1	
	t1	min = 10 max = 4000000 step = 1 unit = ns	*
	t2	Inapplicable	
	r	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
MODULAT	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	r	min = 0 max = 1023 step = 1	
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r0	min = 0 max = r2 step = 1	
	r1	min = 0 max = 1023 step = 1	
	r2	min = r0 max = 1023 step = 1	
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
EOS	m	Inapplicable	
	t	Inapplicable	
	c	Inapplicable	

* The minimum value of the parameter t1 depends on the scanning clock of the e-beam machine. For details, refer to the specification provided at the time of contract. The value min = 10 above is based on the scanning clock being 100 MHz.

Command	Parameter	Limits	Remarks (JBX-9300FS)
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present. It does not work even if you specify this.
SCALEB	bx	Inapplicable	
	by	Inapplicable	
OBJAPT	a	min = 1 max = 8 step = 1	
STDCUR	c	min = 0.001 max = 300.000 step = 0.001 unit = nA	
RESTYP	t	slct = POSI or NEGA	
	n	Up to 20 alphanumeric characters and symbols.	Character string containing single quote (') cannot be specified.
PWIDTH	w	min = 0.010 max = 100.000 step = 0.001 unit = μm	
@	f	Character string consisting of any alphanumeric characters and symbols allowed in the computer system as a filename.	Character string containing single quote (') cannot be specified.

5.2.4 JBX-3030MV

Terminology	
min:	minimum value
max:	maximum value
step:	step
unit:	unit
slct:	selectable value
fix:	fixed value

Table 12 Limits of parameters for JBX-3030MV

Command	Parameter	Limits	Remarks (JBX-3030MV)
JOB	/W	fix = /W	
	j	Up to 64 uppercase letters, numbers, and symbols. Must start with a letter.	Character string containing single quote (') cannot be specified.
	d1	min = 2.0 max = 7.0 step = 0.1 unit= inch	
	d2	min = -7.0 (inches) or -177.800 (mm) max= 7.0 (inches) or 177.800 (mm) step = 0.1 (inches) or 0.001 (mm) unit= inch or mm	
	a	slct = I or M	
	b	slct = I or O	
OPTION	t	fix = T	20-character string that designates date and time: dd-mm-yy, hh:mm:ss.
	i	fix = M	16-character string that designates model name.
	p	slct = X or Y	
	c	slct = X or Y	
	sf	slct = 0 , 1	
DEVINF	d	Up to 64 writable uppercase letters, numbers, and symbols	
GLMPOS	px	min = -88900.000000 max = 88900.000000 step = 0.000001 unit = μ m	
	py	Same as above	
	qx	min = px max = 88900.000000 step = 0.000001 unit = μ m	

Command	Parameter	Limits	Remarks (JBX-3030MV)
	qy	min = -88900.000000 max = 88900.000000 step = 0.000001 unit = μm	
	rx	Inapplicable	
	ry	Inapplicable	
	sx	Inapplicable	
	sy	Inapplicable	
GLMP	w	min = 0.000000 max = 50.000000 step = 0.000001 unit = μm	
	l	min = $w/2$ max = 88900.000000 step = 0.000001 unit = μm	
	t	Inapplicable	
	f	Inapplicable	
GLMQRS	w	min = 0.000000 max = 50.000000 step = 0.000001 unit = μm	
	l	min = $w/2$ max = 88900.000000 step = 0.000001 unit = μm	
	t	Inapplicable	
	f	Inapplicable	
PLMPOS	xn	min = -88900.000000 max = 88900.000000 step = 0.000001 unit = μm	
	yn	Same as above	
PLMARK	w	min = 0.000000 max = 50.000000 step = 0.000001 unit = μm	
	l	min = $w/2$ max = 88900.000000 step = 0.000001 unit = μm	
PATH	/P	fix = /P	
	/S	fix = /S	

Command	Parameter	Limits	Remarks (JBX-3030MV)
	p	Up to 64 letters (uppercase and lowercase letters), numbers, and symbols (period “.” and underscore “_”).	If the character string begins with a character other than a letter, enclose the entire character string in single quotes ('). Character strings used in commands cannot be used.
SCALE	sx	Inapplicable	
	sy	Inapplicable	
ARRAY	a	min = 1 max = 2000 step = 1	
	x	min = -140000.000000 max = 120000.000000 step = 0.000001 unit = μm	*
	m	min = 1 max = 255 step = 1	
	p	min = 0.000000 max = 260000.000000 step = 0.000001 unit = μm	
	y	min = -120000.000000 max = 120000.000000 step = 0.000001 unit = μm	
	n	min = 1 max = 255 step = 1	
	q	min = 0.000000 max = 240000.000000 step = 0.000001 unit = μm	
CHMPOS	x1	min = -260000.000000 max = 260000.000000 step = 0.000001 unit = μm	
	y1	min = -240000.000000 max = 240000.000000 step = 0.000001 unit = μm	

* When the position of a chip array is mirrored with respect to the Y axis according to the specification(s) of the OPTION command in the common block and/or the MIRROR command in a layer block, the limits of the parameter x are changed as follows:
min = -120000.000000, max = 140000.000000, Step = 0.000001, unit = μm .

Command	Parameter	Limits	Remarks (JBX-3030MV)
	x2	min = -260000.000000 max = 260000.000000 step = 0.000001 unit = μm	$x2 - x1 > 0$
	y2	min = -240000.000000 max = 240000.000000 step = 0.000001 unit = μm	
	x3	min = -260000.000000 max = 260000.000000 step = 0.000001 unit = μm	$x3 - x1 > 0$
	y3	min = -240000.000000 max = 240000.000000 step = 0.000001 unit = μm	$y1 - y3 > 0$ & $y2 - y3 > 0$
	x4	min = -260000.000000 max = 260000.000000 step = 0.000001 unit = μm	$x3 - x4 > 0$ & $x2 - x4 > 0$
	y4	min = -240000.000000 max = 240000.000000 step = 0.000001 unit = μm	$y1 - y4 > 0$ & $y2 - y4 > 0$
CHMARK	w	min = 0.000000 max = 50.000000 step = 0.000001 unit = μm	
	l	min = w/2 max = 88900.000000 step = 0.000001 unit = μm	
	t	Inapplicable	
	f	Inapplicable	
ASSIGN	i	min = 1 max = 2000 step = 1	
	a	Same as above	
	j	min = 1 max = 255 step = 1	Asterisk (*) can be used.
	k	Same as above	Same as above
	t	Up to 64 letters (uppercase and lowercase letters), numbers, and symbols (period "." and underscore "_").	If the character string begins with a character other than a letter, enclose the entire character string in single quotes (''). Character strings used in commands cannot be used.

Command	Parameter	Limits	Remarks (JBX-3030MV)
	t1	Inapplicable	
SKIP	j	min = 1 max = 255 step = 1	Asterisk (*) can be used.
	k	Same as above	Same as above
CHAR	i	slct = 1, 2, 3, 4, T, M, D, or L	
	x	min = -140000.000000 max = 120000.000000 step = 0.000001 unit = μm	
	y	min = -120000.000000 max = 120000.000000 step = 0.000001 unit = μm	
	a	slct = 0 or 1	
	b	slct = 0 or 1	
	m	slct = 0, X, or Y	
	r	slct = 0, 90, 180, 270	
P	i	min = 1 max = 2000 step = 1	
	p	A character string can consists of uppercase and lowercase letters, numbers, and symbols. It can contain up to 64 characters, and must be acceptable as a filename in the computer system	The character string can contain up to 256 characters in total, if it includes a directory name (or you can specify the character string using a relative path name). If the character string includes a directory name, the directory must contain the file (do not use dtab). You cannot specify a character string that includes a single quote (').
	δx	min = -260000.000000 max = 260000.000000 step = 0.000001 unit = μm	
	δy	min = -240000.000000 max = 240000.000000 step = 0.000001 unit = μm	
	m	slct = MULTI, ,MULTI/FP GATHER or OVERLAP	You can specify MULTI/FP only in the JBX-3050MV and later models.

Command	Parameter	Limits	Remarks (JBX-3030MV)
	n	min = 2 max = 10 step = 1	Only max = 5 is applicable in m = MULTI, and only max = 8 is applicable in m = OVERLAP.
	csx	min = 0.000000 max = 230000.000000 step = 0.000001 unit = μm	
	csy	Same as above	
SPPRM	m1	min = 0.001 max = 2.000 step = 0.001 unit = μm	*
	m2	min = 0.000001 max = 4.000000 step = 0.000001 unit = μm^2	**
	w	min = 0.001 max = 2.000 step = 0.001 unit = μm	*** The value is limited to an integer multiple (1 to 15 times) of p.
	p	min = 0.001 max = 0.250 step = 0.001 unit = μm	
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	
	sfs	min = 1 max = 5 step = 1	****
STGPRM	a1	slct = XRASTL,XRASTR,XRASBL ,XRASBR,XZIGTL,XZIGTR ,XZIGBL,XZIGBR	*****
	a2	Same as above	
	a3	Same as above	
	a4	Same as above	
	a5	Same as above	

* For the JBX-3050MV, max is limited to 1.000 μm .

** For the JBX-3050MV, max is limited to 1.000000 μm^2 .

*** For the JBX-3050MV, max is limited to 1.000 μm .

**** The feature that enables you to set this parameter is optional for the JBX-3030MV and standard for the JBX-3040MV and later models.

***** The STGPRM command is available only in the JBX-3050MV and later models.

Command	Parameter	Limits	Remarks (JBX-3030MV)
C	i	min = 1 max = 4 step = 1	
	s	Up to 64 uppercase letters, numbers, and symbols that can be written.	*
LAYER	/P	fix = /P	
	n	min = 1 max = 100 step = 1	
	s	Up to 64 uppercase letters, numbers, and symbols that can be written.	
CIRCLE	d2	min = -7.0 (inches) or -177.800 (mm) max = 7.0 (inches) or 177.800 (mm) step = 0.1 (inches) or 0.001 (mm) unit = inch or mm	
	a	slct = I or M	
	b	slct = I or O	
CANCEL	p	slct = T, M, or D	
MIRROR	p	slct = X or Y	
	c	slct = X or Y	
SCALE	sx	Inapplicable	
	sy	Inapplicable	
P	i	min = 1 max = 2000 step = 1	
	p	A character string can consists of uppercase and lowercase letters, numbers, and symbols. It can contain up to 64 characters, and must be acceptable as a filename in the computer system	The character string can contain up to 256 characters in total, if it includes a directory name (or you can specify the character string using a relative path name). If the character string includes a directory name, the directory must contain the file (do not use dtab). You cannot specify a character string that includes a single quote (').
	δx	min = -260000.000000 max = 260000.000000 step = 0.000001 unit = μm	

* The character size is 1.0 mm $\times$ 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-3030MV)
	δy	min = -240000.000000 max = 240000.000000 step = 0.000001 unit = μm	
	m	slct = MULTI, ,MULTI/FP GATHER or OVERLAP	You can specify MULTI/FP only in the JBX-3050MV and later models.
	n	min = 2 max = 10 step = 1	Only max = 5 is applicable in m = MULTI, and only max = 8 is applicable in m = OVERLAP.
	csx	min = 0.000000 max = 230000.000000 step = 0.000001 unit = μm	
	csy	Same as above	
SPPRM	m1	min = 0.001 max = 2.000 step = 0.001 unit = μm	*
	m2	min = 0.000001 max = 4.000000 step = 0.000001 unit = μm^2	**
	w	min = 0.001 max = 2.000 step = 0.001 unit = μm	*** The value is limited to an integer multiple (1 to 15 times) of p.
	p	min = 0.001 max = 0.250 step = 0.001 unit = μm	
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	
	sfs	min = 1 max = 5 step = 1	
	sfs	min = 1 max = 5 step = 1	****

* For the JBX-3050MV, max is limited to 1.000 μm .

** For the JBX-3050MV, max is limited to 1.000000 μm^2 .

*** For the JBX-3050MV, max is limited to 1.000 μm .

**** The feature that allows you to set this parameter is optional for the JBX-3030MV and standard for the JBX-3040MV and later models.

Command	Parameter	Limits	Remarks (JBX-3030MV)
STGPRM	a1	slct = XRASTL,XRASTR,XRASBL ,XRASBR,XZIGTL,XZIGTR ,XZIGBL,XZIGBR	*
	a2	Same as above	
	a3	Same as above	
	a4	Same as above	
	a5	Same as above	
C	i	min = 1 max = 4 step = 1	
	s	Up to 64 uppercase letters, numbers, and symbols that can be written.	**
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	Inapplicable	
	a	Inapplicable	
SHOT	s	Inapplicable	
	t1	Inapplicable	
	t2	Inapplicable	
	r	min = 10.000 max = 150000.000 step = 0.040 unit = ns	
	x	Same as above	
	y	Same as above	
MODULAT	t	Up to 64 letters (uppercase and lowercase letters), numbers, and symbols (period “.” and underscore “_”).	If the character string begins with a character other than a letter, enclose the entire character string in single quotes ('). Character strings used in commands cannot be used.
	r	min = 0 max = 63 step = 1	
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	

* The STGPRM command is available only in the JBX-3050MV and later models.

** The character size is 1.0 mm × 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-3030MV)
	r0	min = 0 max = r2 step = 1	
	r1	min = 0 max = 63 step = 1	
	r2	min = r0 max = 63 step = 1	
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
EOS	m	Inapplicable	
	t	Inapplicable	
	c	Inapplicable	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present. It does not work even if you specify this.
SCALEB	bx	Inapplicable	
	by	Inapplicable	
OBJAPT	a	Inapplicable	
STDCUR	c	Inapplicable	
RESTYP	t	slct = POSI or NEGA	
	n	Up to 64 alphanumeric characters and symbols.	Character string containing single quote (') cannot be specified.
PWIDTH	w	min = 0.010000 max = 100.000000 step = 0.000001 unit = μm	

Command	Parameter	Limits	Remarks (JBX-3030MV)
@	f	A character string can consists of uppercase and lowercase letters, numbers, and symbols. It can contain up to 64 characters, and must be acceptable as a filename in the computer system.	The character string can contain up to 256 characters in total, if it includes a directory name (or you can specify the character string using a relative path name). If the character string includes a directory name, the directory must contain the file. Character string containing single quote (') cannot be specified.

5.2.5 JBX-6000FS/E

Terminology	
min:	minimum value
max:	maximum value
step:	step
unit:	unit
slct:	selectable value
fix:	fixed value

Table 13 Limits of parameters for JBX-6000FS/E

Command	Parameter	Limits	Remarks (JBX-6000FS/E)
JOB	/W	fix = /W	
	j	Up to 9 uppercase letters, numbers, and symbols. Must start with a letter.	Character string containing single quote (') cannot be specified.
	d1	min = 2.0 max = 6.0 step = 0.1 unit= inch	
	d2	min = - 6.0 (inches) or -152.400 (mm) max= 6.0 (inches) or 152.400 (mm) step = 0.1 (inches) or 0.001 (mm) unit= inch or mm	
	a	slct = I or M	
	b	slct = I or O	
OPTION	t	fix = T	15-character string that designates date and time: dd-mmm-yy, hh:mm.
	i	fix = M	16-character string that designates model name.
	p	slct = X or Y	
	c	slct = X or Y	
	sf	Inapplicable	
DEVINF	d	Up to 16 writable uppercase letters, numbers, and symbols	
GLMPOS	px	min = -76200.000 max = 76200.000 step = 0.001 unit = μ m	
	py	Same as above	
	qx	min = px max = 76200.000 step = 0.001 unit = μ m	

Command	Parameter	Limits	Remarks (JBX-6000FS/E)
	qy	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
	rx	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
	ry	min = py max = 76200.000 step = 0.001 unit = μm	
	sx	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
	sy	min = -76200.000 max = py step = 0.001 unit = μm	
GLMP	w	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	l	min = w/2 max = 152400.000 step = 0.001 unit = μm	
	t	slct = 0 or 1	
	f	slct = 0, 1, 2, or 3	
GLMQRS	w	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	l	min = w/2 max = 152400.000 step = 0.001 unit = μm	
	t	slct = 0 or 1	
	f	slct = 0, 1, 2, or 3	
PLMPOS	xn	Inapplicable	
	yn	Inapplicable	
PLMARK	w	Inapplicable	
	l	Inapplicable	
PATH	/P	Inapplicable	

Command	Parameter	Limits	Remarks (JBX-6000FS/E)
	/S	Inapplicable	
	p	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
SCALE	sx	min = 0.970000 max = 1.030000 step = 0.000001	
	sy	Same as above	
ARRAY	a	min = 1 max = 99 step = 1	
	x	min = -104900.000 max = 76200.000 step = 0.001 unit = μm	*
	m	min = 1 max = 255 step = 1	
	p	min = 0.000 max = 152400.000 step = 0.001 unit = μm	
	y	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
	n	min = 1 max = 255 step = 1	
	q	min = 0.000 max = 152400.000 step = 0.001 unit = μm	
CHMPOS	x1	min = -152400.000 max = 152400.000 step = 0.001 unit = μm	
	y1	Same as above	
	x2	Same as above	$x2 \neq x1$
	y2	min = $-(x2-x1) \times \tan 45^\circ + y1$ max = $(x2-x1) \times \tan 45^\circ + y1$ step = 0.001 unit = μm	

* When the position of a chip array is mirrored with respect to the Y axis according to the specification(s) of the OPTION command in the common block and/or the MIRROR command in a layer block, the limits of the parameter x are changed as follows:
min = -76200.000, max = 104900.000, step = 0.001, unit = μm .

Command	Parameter	Limits	Remarks (JBX-6000FS/E)
	x3	$\min = -(y3 - y1) \times \tan 45^\circ + x1$ $\max = (y3 - y1) \times \tan 45^\circ + x1$ step = 0.001 unit = μm	
	y3	min = -152400.000 max = 152400.000 step = 0.001 unit = μm	y3 $\neq$ y1
	X4	Inapplicable	
	Y4	Inapplicable	
CHMARK	w	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	l	min = w/2 max = 152400.000 step = 0.001 unit = μm	
	t	slct = 0 or 1	
	f	slct = 0, 1, 2, or 3	
ASSIGN	i	min = 1 max = 99 step = 1	
	a	Same as above	
	j	min = 1 max = 255 step = 1	Asterisk (*) can be used.
	k	Same as above	Same as above
	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	t1	Inapplicable	
SKIP	j	min = 1 max = 255 step = 1	Asterisk (*) can be used.
	k	Same as above	Same as above
CHAR	i	slct = 1, 2, 3, 4, T, M, D, or L	
	x	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
	y	Same as above	
	a	slct = 0 or 1	
	b	slct = 0 or 1	

Command	Parameter	Limits	Remarks (JBX-6000FS/E)
P	m	Inapplicable	
	r	Inapplicable	
	i	min = 1 max = 99 step = 1	
	p	Up to 24 uppercase letters, numbers, and symbols that can be used as a pattern ID name.	Character string containing single quote (') cannot be specified.
	δx	min = -152400.000 max = 152400.000 step = 0.001 unit = μm	
	δy	Same as above	
	m	Inapplicable	
	n	Inapplicable	
	csx	Inapplicable	
	csy	Inapplicable	
SPPRM	m1	Inapplicable	
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	Inapplicable	
	ss	Inapplicable	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	*
LAYER	/P	Inapplicable	
	n	min = 1 max = 99 step = 1	

* The character size is 1.6 mm $\times$ 1.6 mm, and the writing pitch is 1.6 mm.

Command	Parameter	Limits	Remarks (JBX-6000FS/E)
	s	Up to 16 writable uppercase letters, numbers, and symbols.	
CIRCLE	d2	min = - 6.0 (inches) or - 152.400 (mm) max = 6.0 (inches) or 152.400 (mm) step = 0.1 (inches) or 0.001 (mm) unit = inch or mm	
	a	slct = I or M	
	b	slct = I or O	
CANCEL	p	slct = T, M or D	
MIRROR	p	slct = X or Y	
	c	slct = X or Y	
SCALE	sx	min = 0.970000 max = 1.030000 step = 0.000001	
	sy	Same as above	
P	i	min = 1 max = 99 step = 1	
	p	Up to 24 uppercase letters, numbers, and symbols that can be used as a pattern ID name.	Character string containing single quote (') cannot be specified.
	δx	min = -152400.000 max = 152400.000 step = 0.001 unit = μm	
	δy	Same as above	
	m	Inapplicable	
	n	Inapplicable	
	csx	Inapplicable	
	csy	Inapplicable	
SPPRM	m1	Inapplicable	
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	Inapplicable	
	ss	Inapplicable	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	

Command	Parameter	Limits	Remarks (JBX-6000FS/E)
C	a4	Inapplicable	
	a5	Inapplicable	
	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	*
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	min = 0.001 max = 1000.000 step = 0.001 unit = nC/cm	
	a	Inapplicable	
SHOT	s	min = 1 max = 8 step = 1	**
	t1	min = 167 max = 4000000 step = 1 unit = ns	***
	t2	Same as above	
	r	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
MODULAT	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	r	min = 0 max = 63 step = 1	In single line writing, 16 and up are ignored.
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r0	min = 0 max = r2 step = 1	In single line writing, 16 and up are ignored.

* The character size is 1.6 mm × 1.6 mm, and the writing pitch is 1.6 mm.

** In the 25 kV mode, max = 8 and in the 50 kV mode, max = 16.

*** min = 167 in JBX-5000LS/E.

Command	Parameter	Limits	Remarks (JBX-6000FS/E)
	r1	min = 0 max = 63 step = 1	In single line writing, 16 and up are ignored.
	r2	min = r0 max = 63 step = 1	In single line writing, 16 and up are ignored.
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
EOS	m	min = 1 max = 8 step = 1	
	t	Inapplicable	
	c	A character string of up to 63 characters consisting of printable uppercase and lowercase letters, numbers, and symbols that exclude not only any non-pictorial character including spaces but also slash, backward slash, and single quotation.	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present. It does not work even if you specify this.
SCALEB	bx	min = -0.070000 max = 0.070000 step = 0.000001	
	by	Same as above	
OBJAPT	a	min = 1 max = 3 step = 1	max = 5 in JBX-5DII
STDCUR	c	min = 0.001 max = 100.000 step = 0.001 unit = nA	
RESTYP	t	slct = POSI or NEGA	
	n	Up to 20 alphanumeric characters and symbols.	Character string containing single quote (') cannot be specified.

Command	Parameter	Limits	Remarks (JBX-6000FS/E)
PWIDTH	w	min = 0.010 max = 100.000 step = 0.001 unit = μm	
@	f	Character string consisting of any alphanumeric characters and symbols allowed in the computer system as a filename.	Character string containing single quote (') cannot be specified.

5.2.6 JBX-6300 series

Terminology	
min:	minimum value
max:	maximum value
step:	step
unit:	unit
slct:	selectable value
fix:	fixed value

Table 14 Limits of parameters for JBX-6300 series

Command	Parameter	Limits	Remarks (JBX-6300 series)
JOB	/W	fix = /W	
	j	Up to 9 uppercase letters, numbers, and symbols. Must start with a letter.	Character string containing single quote (') cannot be specified.
	d1	min= 2.0 max= 6.0 step= 0.1 unit= inch	
	d2	min= -6.0 (inch), -152.400(mm) max= 6.0 (inch), 152.400(mm) step= 0.1 (inch), 0.001(mm) unit= inch.mm	
	a	slct= I, or M	
	b	slct= I, or O	
OPTION	t	fix = T	15-character string that designates date and time: dd-mmm-yy, hh:mm.
	i	fix = M	16-character string that designates model name.
	p	slct= X, or Y	
	c	slct= X, or Y	
	sf	Inapplicable	
DEVINF	d	Up to 16 writable uppercase letters, numbers, and symbols	
GLMPOS	px	min= -76200.000 max= 76200.000 step= 0.001 unit= μm	
	py	Same as above	
	qx	min= px max= 76200.000 step= 0.001 unit= μm	
	qy	min= -76200.000 max= 76200.000 step= 0.001 unit= μm	

Command	Parameter	Limits	Remarks (JBX-6300 series)
	rx	min= -76200.000 max= 76200.000 step= 0.001 unit= μm	
	ry	min= -76200.000 max= 76200.000 step= 0.001 unit= μm	
	sx	min= -76200.000 max= 76200.000 step= 0.001 unit= μm	
	sy	min= -76200.000 max= 76200.000 step= 0.001 unit= μm	
GLMP	w	min= 0.000 max= 800.000 step= 0.001 unit= μm	
	l	min= w/2 max= 152400.000 step= 0.001 unit= μm	
	t	slct= 0, or 1	
	f	slct= 0, 1, 2, or 3	
GLMQRS	w	min= 0.000 max= 800.000 step= 0.001 unit= μm	
	l	min= w/2 max= 152400.000 step= 0.001 unit= μm	
	t	slct= 0, or 1	
	f	slct= 0, 1, 2, or 3	
PLMPOS	xn	Inapplicable	
	yn	Inapplicable	
PLMARK	w	Inapplicable	
	l	Inapplicable	
PATH	/P	Inapplicable	
	/S	Inapplicable	
	p	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.

Command	Parameter	Limits	Remarks (JBX-6300 series)
SCALE	sx	min= 0.500000 max= 1.062500 step= 0.000001	When you purchase the fine pitch control writing function: min= 0.937500 max= 1.062500 step= 0.000001 When you purchase the minimum increment variable writing function; min= 0.500000 max= 1.000000 step= 0.125000
	sy	Same as above	
ARRAY	a	min= 1 max= 99 step= 1	
	x	min= -104900.000 max= 76200.000 step= 0.001 unit= μm	*
	m	min= 1 max= 255 step= 1	
	p	min= 0.000 max= 152400.000 step= 0.001 unit= μm	
	y	min= -76200.000 max= 76200.000 step= 0.001 unit= μm	
	n	min= 1 max= 255 step= 1	
	q	min= 0.000 max= 152400.000 step= 0.001 unit= μm	
CHMPOS (In the case of 3 chip marks)	x1	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	
	y1	Same as above	
	x2	Same as above	$x2 \neq x1$
	y2	min = $-(x2 - x1) \times \tan 45^\circ + y1$ max = $(x2 - x1) \times \tan 45^\circ + y1$ step= 0.001 unit= μm	

* When the position of a chip array is mirrored with respect to the Y axis according to the specification(s) of the OPTION command in the common block and/or the MIRROR command in a layer block, the limits of the parameter x are changed as follows:
min = -76200.000, max = 104900.000, step = 0.001, unit = μm .

Command	Parameter	Limits	Remarks (JBX-6300 series)
	x3	$\min = -\left(\left y_3 - x_1\right \right) \times \tan 45^\circ + x_1$ $\max = \left(\left y_3 - x_1\right \right) \times \tan 45^\circ + x_1$ step= 0.001 unit= μm	
	y3	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	$y_3 \neq y_1$
	x4	Inapplicable	
	y4	Inapplicable	
CHMPOS (In the case of 3 chip marks)	x1	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	
	y1	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	
	x2	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	$x_2 - x_1 > 0$
	y2	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	
	x3	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	$x_3 - x_1 > 0$
	y3	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	$y_1 - y_3 > 0$ and $y_2 - y_3 > 0$
	x4	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	$x_3 - x_4 > 0$ and $x_2 - x_4 > 0$
	y4	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	$y_1 - y_4 > 0$ and $y_2 - y_4 > 0$
CHMARK	w	min= 0.000 max= 800.000 step= 0.001 unit= μm	
	l	min= $w/2$ max= 152400.000 step= 0.001 unit= μm	
	t	slct=0, or 1	
	f	slct=0, 1, 2, or 3	

Command	Parameter	Limits	Remarks (JBX-6300 series)
ASSIGN	i	min= 1 max= 99 step= 1	
	a	Same as above	
	j	min= 1 max= 255 step= 1	Asterisk (*) can be used.
	k	Same as above	Same as above
	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	tl	Up to 6 uppercase letters and numbers. Must start with a letter.	The name of shot time modulation table for single line data is specified. If omitted, shot time modulation table for tl is used. Character strings used in commands cannot be specified.
SKIP	j	min= 1 max= 255 step= 1	Asterisk (*) can be used.
	k	Same as above	Same as above
CHAR	i	slct=1, 2, 3, 4, T, M, D, or L	
	x	min= -76200.000 max= 76200.000 step= 0.001 unit= μm	
	y	Same as above	
	a	slct=0, or 1	
	b	slct=0, or 1	
	m	Inapplicable	
	r	Inapplicable	
P	i	min= 1 max= 99 step= 1	
	p	Up to 24 uppercase letters, numbers, and symbols that can be used as a pattern ID name.	Character string containing single quote (') cannot be specified.
	δx	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	
	δy	Same as above	
	m	slct=MULTI,GATHER,OVERLAP	

Command	Parameter	Limits	Remarks (JBX-6300 series)
	n	min= 2 max= 10 step= 1	In the case of m = MULTI or m = OVERLAP, max = 5 or max = 8, respectively.
	csx	Inapplicable	
	csy	Inapplicable	
SPPRM	m1	min= 0.4 max= 4.000 step= 0.002 unit= μm	For EOS mode 2.*
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	min= 0.500000 max= 1.200000 step= 0.000001	
	ss	min= 1 max= 5 step= 1	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min= 1 max= 4 step= 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	**

* The limits depend on the EOS mode. This table shows the limits in each mode.

EOS mode	min	max	step
1 (25 kV-4th lens mode)	0.8	8.000	0.004
2 (50 kV-4th lens mode)	0.4	4.000	0.002
3 (100 kV-4th lens mode)	0.2	2.000	0.001
4 (25 kV-5th lens mode)	0.1	1.000	0.001
5 (50 kV-5th lens mode)	0.05	0.500	0.001
6 (100 kV-5th lens mode)	0.025	0.250	0.001

** The character size is 1.0 mm $\times$ 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-6300 series)
LAYER	/P	Inapplicable	
	n	min= 1 max= 99 step= 1	
	s	Up to 16 writable uppercase letters, numbers, and symbols.	
CIRCLE	d2	min= -6.0(inch), -152.400(mm) max= 6.0(inch), 152.400(mm) step= 0.1(inch), 0.001(mm) unit= inch,mm	
CIRCLE	a	slct=I, or M	
	b	slct=I, or O	
CANCEL	p	slct=T, M, or D	
MIRROR	p	slct=X, or Y	
	c	slct=X, or Y	
SCALE	sx	min= 0.500000 max= 1.062500 step= 0.000001	When you purchase the fine pitch control writing function: min= 0.937500 max= 1.062500 step= 0.000001 When you purchase the minimum increment variable writing function; min= 0.500000 max= 1.000000 step= 0.125000
	sy	Same as above	
P	i	min= 1 max= 99 step= 1	
	p	Up to 24 uppercase letters, numbers, and symbols that can be used as a pattern ID name.	Character string containing single quote (') cannot be specified.
	δx	min= -152400.000 max= 152400.000 step= 0.001 unit= μm	
	δy	Same as above	
	m	slct=MULTI,GATHER,OVERLAP	
	n	min= 2 max= 10 step= 1	In the case of m = MULTI or m = OVERLAP, max = 5 or max = 8, respectively.
	csx	Inapplicable	
	csy	Inapplicable	

Command	Parameter	Limits	Remarks (JBX-6300 series)
SPPRM	m1	min= 0.4 max= 4.000 step= 0.002 unit= μm	For EOS mode 2.*
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	min= 0.500000 max= 1.200000 step= 0.000001	
	ss	min= 1 max= 5 step= 1	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min= 1 max= 4 step= 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	*
RESIST	s1	min= 0.010 max= 1000000.000 step= 0.001 unit= $\mu\text{C}/\text{cm}^2$	

* The limits depend on the EOS mode. This table shows the limits in each mode.

EOS mode	min	max	step
1 (25 kV-4th lens mode)	0.8	8.000	0.004
2 (50 kV-4th lens mode)	0.4	4.000	0.002
3 (100 kV-4th lens mode)	0.2	2.000	0.001
4 (25 kV-5th lens mode)	0.1	1.000	0.001
5 (50 kV-5th lens mode)	0.05	0.500	0.001
6 (100 kV-5th lens mode)	0.025	0.250	0.001

* The character size is 1.0 mm $\times$ 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-6300 series)
	s2	a= A ; Area sensitivity min= 0.010 max= 1000000.000 step= 0.001 unit = $\mu\text{C}/\text{cm}^2$ a= L ; Line sensitivity min= 0.001 max= 1000000.000 step= 0.001 unit= nC/cm	
	a	A= $\mu\text{C}/\text{cm}^2$ L= nC/cm	
FSHOT	s	min= 1 max= 254 step= 1	
	t1	min= 84 max= 4000000 step= 1 unit= ns	*
	t2	Same as above	
	r	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
MODULAT	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	r	min= 0 max= 254 step= 1	For single-line data, max = 15.
	v	min= -99.9 max= 3200.0 step= 0.1 unit= %	
	r0	min= 0 max= r2 step= 1	
	r1	min= 0 max= 254 step= 1	For single-line data, max = 15.
	r2	min= r0 max= 254 step= 1	For single-line data, max = 15.
	v0	min= -99.9 max= v2 step= 0.1 unit= %	
	v1	min= -50.0 max= 100.0 step= 0.1 unit= %	

* The value of “min” depends on the scan clock for the lithography system. For more details, refer to the instruction manual for the lithography system. Here, the value of “min” is the value under the assumption that the scan clock = 12 MHz.

Command	Parameter	Limits	Remarks (JBX-6300 series)
	v2	Min= v0 max= 3200.0 step= 0.1 unit= %	
EOS	m	Min= 1 max= 6 step= 1	
	t	Inapplicable	
	c	A character string of up to 63 characters consisting of printable uppercase and lowercase letters, numbers, and symbols that exclude not only any non-pictorial character including spaces but also slash, backward slash, and single quotation.	
DEFOCUS	d	min= -4095 max= 4095 step= 1	Inapplicable at present. It does not work even if you specify this.
SCALEB	bx	Inapplicable	
	by	Inapplicable	
OBJAPT	a	min= 1 max= 8 step= 1	
STDCUR	c	min= 0.001 max= 100.000 step= 0.001 unit= nA	
RESTYP	t	slct= POSI, or NEGA	
	n	Up to 20 alphanumeric characters and symbols.	Character string containing single quote (') cannot be specified.
PWIDTH	w	min= 0.010 max= 100.000 step= 0.001 unit= μm	
@	f	Character string consisting of any alphanumeric characters and symbols allowed in the computer system as a filename.	Character string containing single quote (') cannot be specified.

5.2.7 JBX-5500 series

Terminology	
min:	minimum value
max:	maximum value
step:	step
unit:	unit
slct:	selectable value
fix:	fixed value

Table 15 Limits of parameters for JBX-5500 series

Command	Parameter	Limits	Remarks (JBX-5500 series)
JOB	/W	fix = /W	
	j	Up to 9 uppercase letters, numbers, and symbols. Must start with a letter.	Character string containing single quote (') cannot be specified.
	d1	min= 2.0 max= 4.0 step= 0.1 unit= inch	
	d2	min= -4.0(inch), -101.600(mm) max= 4.0(inch), 101.600(mm) step= 0.1(inch), 0.001(mm) unit= inch, mm	
	a	slct= I, or M	
	b	slct= I, or O	
OPTION	t	fix= T	15-character string that designates date and time: dd-mmm-yy, hh:mm.
	i	fix= M	16-character string that designates model name.
	p	slct= X, or Y	
	c	slct= X, or Y	
	sf	Inapplicable	
DEVINF	d	Up to 16 writable uppercase letters, numbers, and symbols	
GLMPOS	px	min= -50800.000 max= 50800.000 step= 0.001 unit= μm	
	py	Same as above	
	qx	min= px max= 50800.000 step= 0.001 unit= μm	

Command	Parameter	Limits	Remarks (JBX-5500 series)
	qy	min= -50800.000 max= 50800.000 step= 0.001 unit= μm	
	rx	Inapplicable	
	ry	Inapplicable	
	sx	Inapplicable	
	sy	Inapplicable	
GLMP	w	min= 0.000 max= 800.000 step= 0.001 unit= μm	
	l	min= w/2 max= 101600.000 step= 0.001 unit= μm	
	t	slct = 0, or 1	
	f	slct = 0, 1, 2, or 3	
GLMQRS	w	min= 0.000 max= 800.000 step= 0.001 unit= μm	
	l	min= w/2 max= 101600.000 step= 0.001 unit= μm	
	t	slct= 0, or 1	
	f	slct= 0, 1, 2, or 3	
PLMPOS	xn	Inapplicable	
	yn	Inapplicable	
PLMARK	w	Inapplicable	
	l	Inapplicable	
PATH	/P	Inapplicable	
	/S	Inapplicable	
	p	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
SCALE	sx	min= 0.940000 max= 1.060000 step= 0.000001	With the purchase of the fine pitch control writing function
	sy	Same as above	Same as above

Command	Parameter	Limits	Remarks (JBX-5500 series)
ARRAY	a	min= 1 max= 2000 step= 1	
	x	min= -79500.000 max= 50800.000 step= 0.001 unit= μm	*
	m	min= 1 max= 255 step= 1	
	p	min= 0.000 max= 101600.000 step= 0.001 unit= μm	
	y	min= -50800.000 max= 50800.000 step= 0.001 unit= μm	
	n	min= 1 max= 255 step= 1	
	q	min= 0.000 max= 101600.000 step= 0.001 unit= μm	
CHMPOS	x1	min= -101600.000 max= 101600.000 step= 0.001 unit= μm	
	y1	Same as above	
	x2	Same as above	$x2 \neq x1$
	y2	min= $-(x2-x1) \times \tan 45^\circ + y1$ max= $(x2-x1) \times \tan 45^\circ + y1$ step= 0.001 unit= μm	
	x3	min= $-(y3-y1) \times \tan 45^\circ + x1$ max= $(y3-y1) \times \tan 45^\circ + x1$ step= 0.001 unit= μm	
	y3	min= -101600.000 max= 101600.000 step= 0.001 unit= μm	$y3 \neq y1$
	x4	Inapplicable	

* When the position of a chip array is mirrored with respect to the Y axis according to the specification(s) of the OPTION command in the common block and/or the MIRROR command in a layer block, the limits of the parameter x are changed as follows:
min = -50800.000, max = 79500.000, step = 0.001, unit = μm .

Command	Parameter	Limits	Remarks (JBX-5500 series)
	y4	Inapplicable	
CHMARK	w	min= 0.000 max= 800.000 step= 0.001 unit= μm	
	l	min= w/2 max= 101600.000 step= 0.001 unit= μm	
	t	slct=0, or 1	
	f	slct=0, 1, 2, or 3	
ASSIGN	i	min= 1 max= 99 step= 1	
	a	Same as above	
	j	min= 1 max= 255 step= 1	Asterisk (*) can be used.
	k	Same as above	Same as above
	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	tl	Inapplicable	
SKIP	j	min= 1 max= 255 step= 1	Asterisk (*) can be used.
	k	Same as above	Same as above
CHAR	i	slct= 1,2 ,3, 4, T, M, D, or L	
	x	min= -50800.000 max= 50800.000 step= 0.001 unit= μm	
	y	Same as above	
	a	slct= 0, or 1	
	b	slct= 0, or 1	
	m	Same as above	
	r	Same as above	
P	i	min= 1 max= 99 step= 1	
	p	Up to 24 uppercase letters, numbers, and symbols that can be used as a pattern ID name.	Character string containing single quote (') cannot be specified.
	δx	min= -101600.000 max= 101600.000 step= 0.001 unit= μm	

Command	Parameter	Limits	Remarks (JBX-5500 series)
	δy	Same as above	
	m	slct=MULTI, GATHER, OVERLAP	
	n	min= 2 max= 10 step= 1	In the case of m = MULTI or m = OVERLAP, max = 5 or max = 8, respectively.
	csx	Inapplicable	
	csy	Inapplicable	
SPPRM	m1	min= 1.000 max= 10.000 step= 0.005 unit= μm	For EOS mode 2.*
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	min= 0.500000 max= 1.200000 step= 0.000001	
	ss	min= 1 max= 5 step= 1	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min= 1 max= 4 step= 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	**

* The limits depend on the EOS mode. This table shows the limits in each mode.

EOS mode	min	max	step
1 (25 kV-4th lens mode)	2.000	20.000	0.010
2 (50 kV-4th lens mode)	1.000	10.000	0.005
3 (25 kV-5th lens mode)	0.200	2.000	0.001
4 (50 kV-5th lens mode)	0.100	1.000	0.001

** The character size is 1.0 mm $\times$ 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-5500 series)
LAYER	/P	Inapplicable	
	n	min= 1 max= 99 step= 1	
	s	Up to 16 writable uppercase letters, numbers, and symbols.	
CIRCLE	d2	min= -4.0(inch), -101.600(mm) max= 4.0(inch), 101.600(mm) step= 0.1(inch), 0.001(mm) unit= inch, mm	
	a	slct= I, or M	
	b	slct= I, or O	
CANCEL	p	slct= T, M, or D	
MIRROR	p	slct= X, or Y	
	c	slct= X, or Y	
SCALE	sx	min= 0.940000 max= 1.060000 step= 0.000001	With the purchase of the fine pitch control writing function
	sy	Same as above	
P	i	min= 1 max= 99 step= 1	
	p	Up to 24 uppercase letters, numbers, and symbols that can be used as a pattern ID name.	Character string containing single quote (') cannot be specified.
	δx	min= -101600.000 max= 101600.000 step= 0.001 unit= μm	
	δy	Same as above	
	m	slct= MULTI, GATHER, OVERLAP	
	n	min= 2 max= 10 step= 1	In the case of m = MULTI or m = OVERLAP, max = 5 or max = 8, respectively.
	csx	Inapplicable	
	csy	Inapplicable	

Command	Parameter	Limits	Remarks (JBX-5500 series)
SPPRM	m1	min= 1.000 max= 10.000 step= 0.005 unit= μm	For EOS mode 2.*
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	min= 0.500000 max= 1.200000 step= 0.000001	
	ss	min= 1 max= 5 step= 1	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min= 1 max= 5 step= 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	**
RESIST	s1	min= 0.010 max= 1000000.000 step= 0.001 unit= $\mu\text{C}/\text{cm}^2$	

* The limits depend on the EOS mode. This table shows the limits in each mode.

EOS mode	min	max	step
1 (25 kV-4th lens mode)	2.000	20.000	0.010
2 (50 kV-4th lens mode)	1.000	10.000	0.005
3 (25 kV-5th lens mode)	0.200	2.000	0.001
4 (50 kV-5th lens mode)	0.100	1.000	0.001

** The character size is 1.0 mm $\times$ 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-5500 series)
	s2	a= A ; Area sensitivity min= 0.010 max= 1000000.000 step= 0.001 unit = $\mu\text{C}/\text{cm}^2$ a= L ; Line sensitivity min= 0.001 max= 1000000.000 step= 0.001 unit= nC/cm	
	a	A= $\mu\text{C}/\text{cm}^2$ L= nC/cm	
SHOT	s	min= 1 max= 254 step= 1	
	t1	min= 84 max= 4000000 step= 1 unit= ns	*
	t2	Same as above	
	r	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
MODULAT	t	Up to 6 uppercase letters and numbers. Must start with a letter.	Character strings used in commands cannot be used.
	r	min= 0 max= 255 step= 1	In single line writing, 16 and up are ignored.
	v	min= -99.9 max= 3200.0 step= 0.1 unit= %	
	r0	min= 0 max= r2 step= 1	
	r1	min= 0 max= 255 step= 1	In single line writing, 16 and up are ignored.
	r2	min= r0 max= 255 step= 1	In single line writing, 16 and up are ignored.
	v0	min= -99.9 max= v2 step= 0.1 unit= %	

* The value of “min” depends on the scan clock for the lithography system. For more details, refer to the instruction manual for the lithography system. Here, the value of “min” is the value under the assumption that the scan clock = 12 MHz.

Command	Parameter	Limits	Remarks (JBX-5500 series)
	v1	min= -50.0 max= 100.0 step= 0.1 unit= %	
	v2	min= v0 max= 3200.0 step= 0.1 unit= %	
EOS	m	min= 1 max= 4 step= 1	
	t	Inapplicable	
	c	A character string of up to 63 characters consisting of printable uppercase and lowercase letters, numbers, and symbols that exclude not only any non-pictorial character including spaces but also slash, backward slash, and single quotation.	
DEFOCUS	d	min= -4095 max= 4095 step= 1	Inapplicable at present. It does not work even if you specify this.
SCALEB	bx	Inapplicable	
	by	Inapplicable	
OBJAPT	a	min= 1 max= 4 step= 1	
STDCUR	c	min = 0.001 max = 100.000 step = 0.001 unit = nA	
RESTYP	t	slct = POSI or NEGA	
	n	Up to 20 alphanumeric characters and symbols.	Character string containing single quote (') cannot be specified.
PWIDTH	w	min = 0.010 max = 100.000 step = 0.001 unit = μm	
@	f	Character string consisting of any alphanumeric characters and symbols allowed in the computer system as a filename.	Character string containing single quote (') cannot be specified.

5.3 Writable Characters

Table 16 Writable characters

Character	Code	Character	Code
(SPACE)	040	@	100
!	041	A	101
”	042	B	102
#	043	C	103
		D	104
%	045	E	105
&	046	F	106
		G	107
		H	110
		I	111
*	052	J	112
+	053	K	113
,	054	L	114
–	055	M	115
.	056	N	116
/	057	O	117
0	060	P	120
1	061	Q	121
2	062	R	122
3	063	S	123
4	064	T	124
5	065	U	125
6	066	V	126
7	067	W	127
8	070	X	130
9	071	Y	131
:	072	Z	132
;	073	[	133
<	074	\	134
=	075	]	135
>	076	^	136
?	077	–	137

5.4 Examples of Jobdeck Files

5.4.1 Jobdeck for direct writing (Example A)

The following example is for JBX-5000LS, JBX-6000FS, etc.

```
JOB/W      ' DIRECT01 ', 3, -3
;
  GLMPOS      P=(-35000, 0), Q=(35000, 0), R=(0, 35000), S=(0, -35000)
  GLMP        3. 0, 3200. 0, 0, 0
  GLMQRS      3. 0, 3200. 0, 0, 0
;
  PATH        JEOL01
  ARRAY       (-30000, 13, 5000) / (30000, 13, 5000)
  CHMPOS      M1=(-2000, 2000), M2=(-500, 2000), M3=(-2000, 500)
  CHM123      3. 0, 20. 0, 1, 0
  ASSIGN      P(1) -> ((*,*), SHOT01)
  AEND
  PEND
;
  LAYER       1
  P(1)        ' TEST001 '
  EOS         1, 0
; mode=1, table=0

  SHOT        A, 2
  RESIST      600, 3
  OBJAPT      2 ; No. 2
  STDCUR      5 ; 5 nA
  RESTYP      POSI, ' PMMA '
  PWIDTH      0. 10 ; 0. 10  $\mu$ m
  SHOT01: MODULAT ((0, -10))
;
END
```

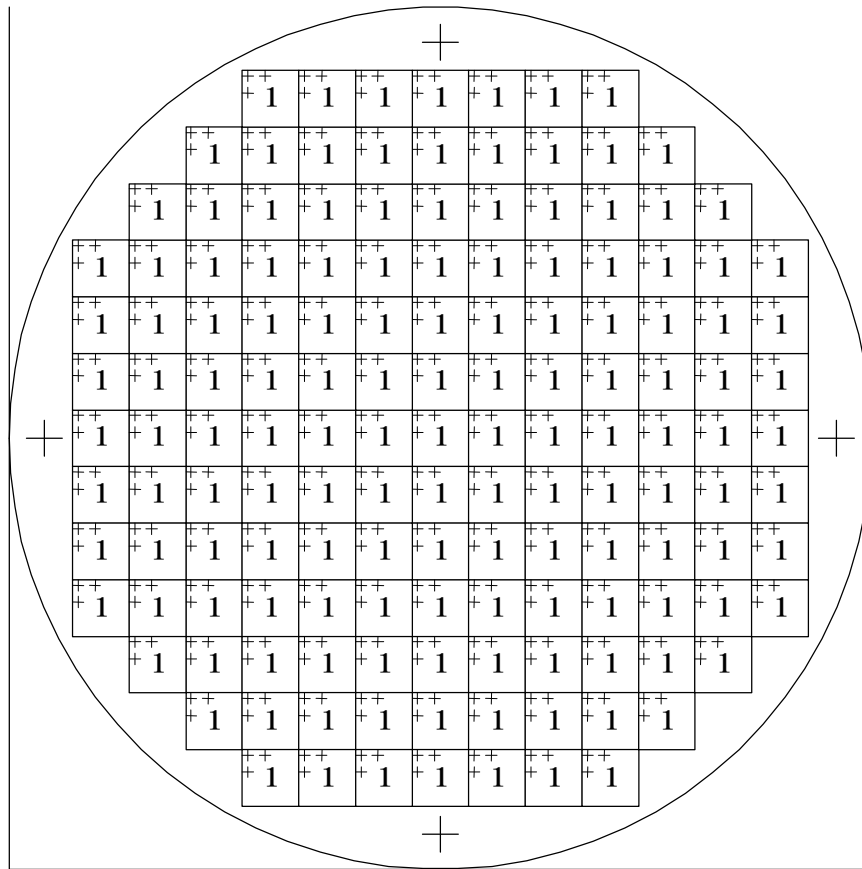


Fig.2 Chip array pattern for direct writing (Example A)

5.4.2 Jobdeck for direct writing (Example B)

The following example is for JBX-9300FS.

```
JOB/W      ' DIRECT02 ', 12
;
GLMPOS      P=(-110000, 0), Q=(110000, 0)
GLMP        3. 0, 3200. 0, 0, 0
GLMQRS      3. 0, 3200. 0, 0, 0
;
      PATH      CAL001
      ARRAY      (-96000, 17, 12000) / (96000, 17, 12000)
      CHMPOS      M1=(-6000, 6000), M2=(6000, 6000), M3=(6000, -6000),
      -          M4=(-6000, -6000)
      CHMARK      3. 0, 20. 0, 1, 0
      ASSIGN      P(1) + P(2) -> ((*,*), SHOT01)
      AEND
      PEND
;
      LAYER      1
      P(1)      ' GATE001 ' (-3000, 0), OVERLAP, 2
      SPPRM      4. 0, , , , 2
      P(2)      ' GATE002 ' (3000, 0), OVERLAP, 2
      SPPRM      4. 0, , , , 2
      SHOT      A, 2
      RESIST      600
      OBJAPT      4                      ; No. 4
      STDCUR      5                      ; 5 nA
      RESTYP      POSI, ' PMMA '
      PWIDTH      0. 10                  ; 0. 10 um
SHOT01: MODULAT ((0, -10))
;
END
```

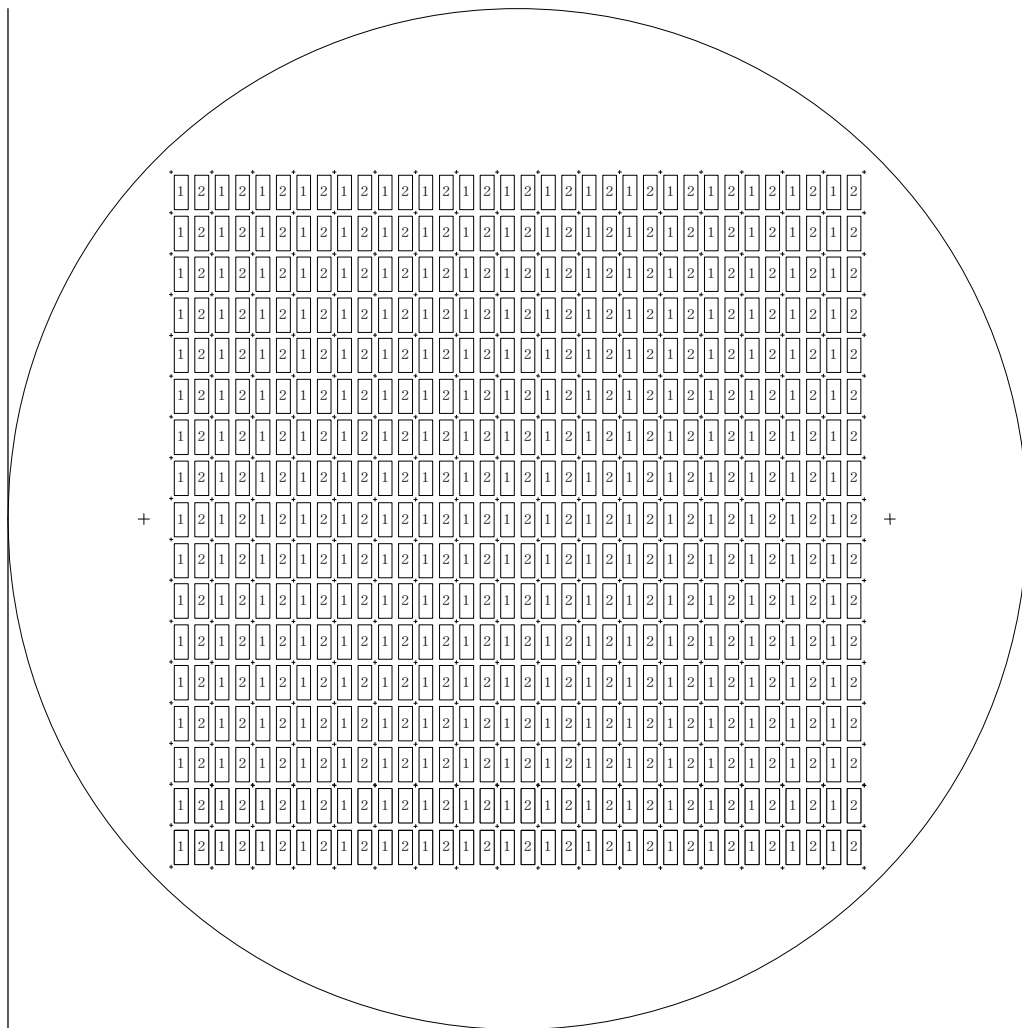


Fig.3 Chip array pattern for direct writing (Example B)

5.4.3 Jobdeck for mask pattern writing

The following example is for JBX-6000FS.

```
JOB      ' MASK01 ' , 3, -2.9
;
PATH      JEOL01
ARRAY      ( -20000, 4, 5000 ) / ( 25000, 5, 5000 )
  ASSIGN    P( 1) -> ( (*, *), SHOT01 )
  ASSIGN    P(99) -> ( (2, 4), (3, 5), SHOT99 )
  SKIP      ( (3, 2-3) )
AEND
CHAR/1 (0.0, 30000.0)
PEND
PATH JEOL01
ARRAY      ( 5000, 4, 5000 ) / ( 25000, 5, 5000 )
  ASSIGN    P( 2) -> ( (*, *), SHOT02 )
AEND
PEND
PATH JEOL01
ARRAY      ( -35000, 7, 5000 ) / ( -5000, 7, 5000 )
  ASSIGN    P( 3) -> ( (*, *), SHOT03 )
AEND
PEND
PATH JEOL01
ARRAY      ( 5000, 10, 5000 ) / ( -5000, 10, 5000 )
  ASSIGN    P( 4) -> ( (*, *), SHOT04 )
  SKIP      ( (3-4, 2) )
AEND
PEND
;
LAYER      1
  P( 1)      ' TEST001 '
  P( 2)      ' TEST002 '
  P( 3)      ' TEST003 '
  P( 4)      ' TEST004 '
  P(99)      ' TEST099 '
  C( 1)      ' TEST '
  EOS        1, 0 ; mode=1, table=0
  SHOT       A, 2
  RESIST     200, 100
  OBJAPT     2 ; No. 2
  STDCUR     5 ; 5 nA
  RESTYP     POSI, ' PMMA '
  PWIDTH     0.10 ; 0.10 um
SHOT01: MODULAT ( ( 0, -10 ) )
SHOT02: MODULAT ( ( 0, -20 ) )
SHOT03: MODULAT ( ( 0, -30 ) )
SHOT04: MODULAT ( ( 0, -40 ) )
```



```
SHOT99: MODULAT ( ( 0,0  ) )  
;  
END
```

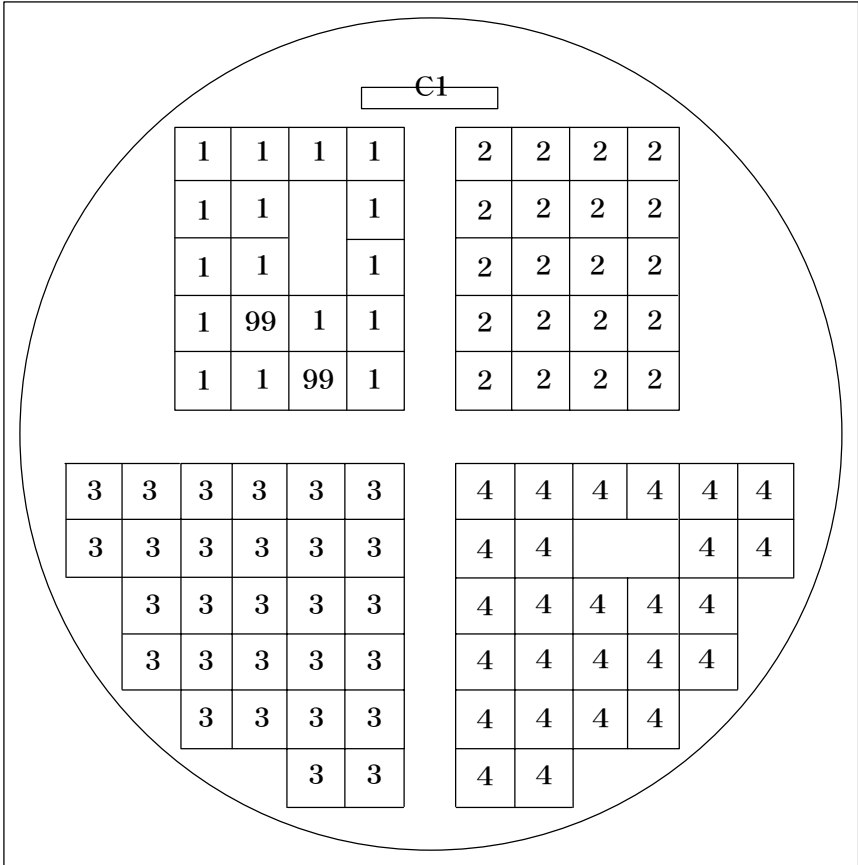


Fig.4 Chip array pattern for mask pattern writing

5.4.4 Jobdeck having chip array definitions using ASSIGN command

The following example is for JBX-9000MV and JBX-3030MV.

```
JOB    'BLOCK01 ', 6, 0
;
    PATH    JEOL01
    ARRAY    ( -35000, 2, 70000 ) / ( 35000, 2, 70000 )
        ASSIGN    P(1) + P(2) + A(1) -> (1, 1)
        ASSIGN    P(1) + P(3) + A(2) -> (2, 1)
        ASSIGN    P(1) + P(4) + A(1) -> (1, 2)
        ASSIGN    P(1) + P(5) + A(2) -> (2, 2)
    AEND
1:      ARRAY    ( 0, 3, 10000 ) / ( 20000, 2, 10000 )
        ASSIGN    P(6) -> (*, *)
        AEND
2:      ARRAY    ( 10000, 2, 10000 ) / ( 20000, 5, 10000 )
        ASSIGN    P(6) -> (*, *)
        AEND
    PEND
;
    LAYER      1
        P(1)    'test001.v30 '
        P(2)    'test002.v30 ' (-15000, -15000)
        P(3)    'test003.v30 ' (-15000, -15000)
        P(4)    'test004.v30 ' ( 15000, -15000)
        P(5)    'test005.v30 ' (-15000, -15000)
        P(6)    'test006.v30 '
        RESIST  2.4
;
END
```

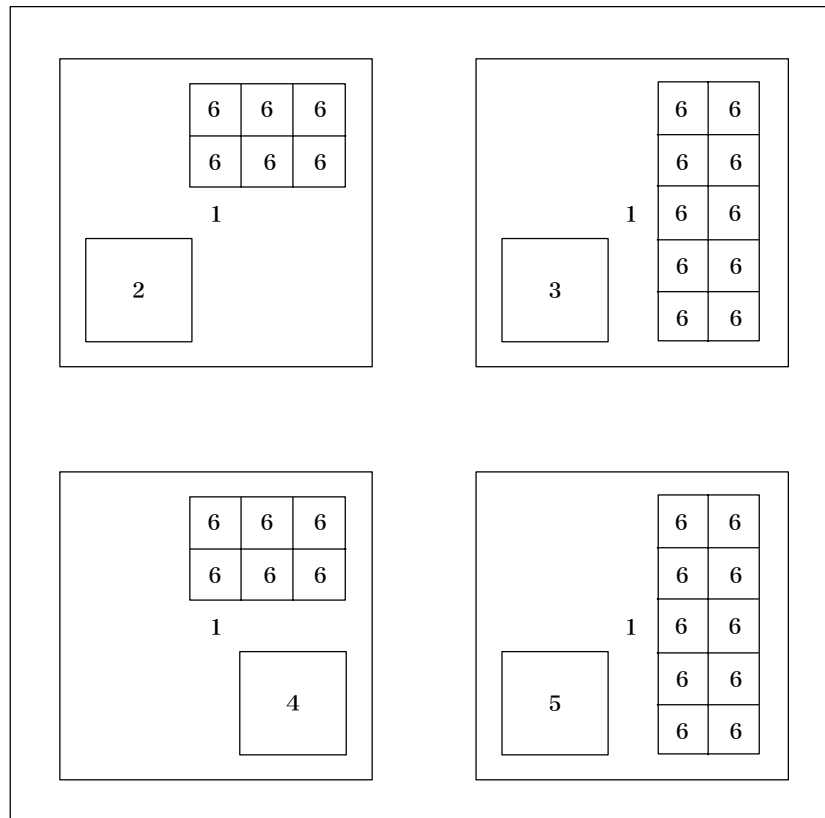


Fig.5 Chip array pattern allotted by ASSIGN command

5.4.5 Jobdeck for phase shift mask pattern writing

The following example is for JBX-9000MV and JBX-3030MV.

```

JOB      ' PHP01 ' , 6, 0
;
  GLMPOS P=(-65000, 0), Q=(65000, 0)
  GLMP      10. 0, 750. 0
  PLMPOS (-65000, 0), (65000, 0), (0, 65000), (0, -65000),
-          (65000, 65000), (-65000, -65000),
-          (-65000, 65000), (65000, -65000), (0, 0)
  PLMARK 10. 0, 40. 0
;
  PATH      MARK01
  ARRAY      ( -65000, 3, 65000 )/( 65000, 3, 65000 )
  ASSIGN      P( 2) -> (*, *)
  ASSIGN      P( 3) -> ((1, 2), (3, 2))
  AEND
  PEND
  PATH      JEOL01
  ARRAY      ( -32500, 2, 65000 )/( 0, 1, 0 )
  ASSIGN      P( 1) -> (1, 1)
  ASSIGN      P(11) -> (2, 1)
  AEND
  PEND
;
  LAYER      1
  P( 1)      ' device001.v30 ' , MULTI, 2
  P(11)      ' device011.v30 ' , MULTI, 2
  SPPRM      2. 0, 4. 0, 0. 24, 0. 12, 0. 75, 2
  P( 2)      ' platemark.v30 '
  P( 3)      ' globalmark.v30 '
  LAYER/P    2
  P( 1)      ' device001shift.v30 ' , MULTI, 2
  P(11)      ' device011shift.v30 ' , MULTI, 2
  SPPRM      2. 0, 4. 0, 0. 24, 0. 12, 0. 75, 2
;
END

```

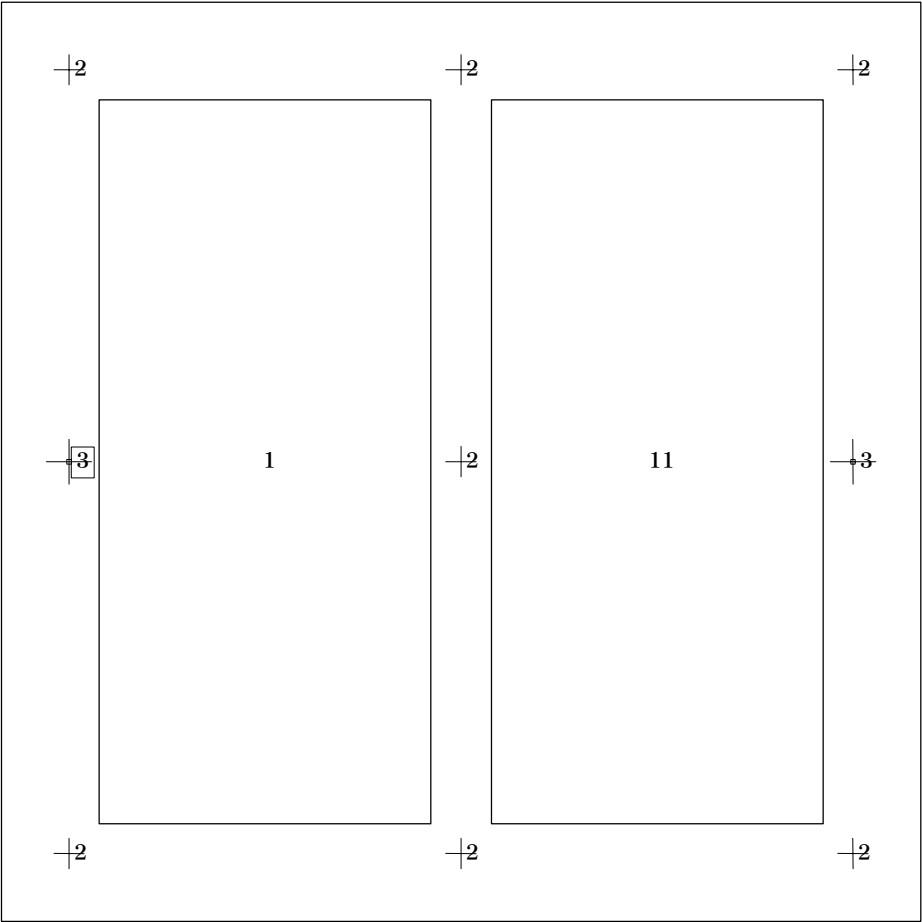


Fig.6 Chip array pattern for phase shift mask pattern writing